# On the least prime in an arithmetic progression

## I. The basic theorem

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In the paper entitled «On the possibility of avoiding Riemann's extended hypothesis in the studies on primes in progressions» [1] I pointed out that certain «qualitative problems» of the arithmetics of primes can be resolved, if it is known that any $L$-series $L(s, \chi)$ to a modulus $D$ possesses the following «density property»:

(d) *Any circle with the radius* $\frac{1}{\ln D}$ *and the center in the rectangle*

$$1 \geqslant \sigma \geqslant 0{,}9; \quad |t| \leqslant \ln^3 D$$

*contains no more than* $O(1)$ *zeros of* $L(s, \chi)$ *for* $D \to \infty$.

The arguments outlined in the paper [1] are sufficient even to give a more general result than is stated there (the result about «one half of progressions»). Namely, if (d) holds for any $L(s, \chi)$ (mod $D$) and $p_{\min}(l, D)$ is the least prime in an arithmetic progression $Dx+l$ with $l \in [1, D-1]$ and $(l, D)=1$, then

$$\lim_{D\to\infty} \frac{\ln p_{\min}(l, D)}{\ln D} < \infty. \tag{1}$$

This is equivalent to the assertion that there exists an absolute constant $C_0$ such that

$$p_{\min}(l, D) \leqslant D^{C_0}. \tag{2}$$

The object of the present paper is to prove (1) and (2) without any hypothesis.

The paper is divided into three sections: I — «The basic theorem»; II — «Deuring — Heilbronn phenomenon»; III — «The weak asymptotic law».

The gist of these sections is the following: in I is proved the «basic theorem» on the quantity of the series $L(s, \chi)$ with the zeros in certain rectangles, which is very similar, but weaker than the crucial estimate (8) of the paper [1], which was proved there only conditionally, under the property (d). In II the well known difficulties connected with the real zeros are treated on the ground of Deuring — Heilbronn's ideas, but by a new method. After that, the obtained results are applied to the proof of (1) and (2). In III a new weak asymptotic law is deduced.

The knowledge of my papers: [1]; «On the distribution of characters» [2]; «On the characters of primes, I» [3], «On Dirichlet's $L$-series and prime number sums» [4] is assumed.

## § 1. The basic theorem

Let $L(s, \chi_1)$, $L(s, \chi_2)$, ..., $L(s, \chi_{\varphi(D)})$ be all the series to a modulus $D$. Let $\Psi(D)$ be any number under conditions $\frac{1}{3} \ln D \geqslant \Psi(D) \geqslant 2$.

Let $Q\left(\frac{\Psi(D)}{\ln D}\right)$ be the quantity of $L$-series, each having at least one zero in the rectangle

$$1 \geqslant \sigma \geqslant 1 - \frac{\Psi(D)}{\ln D}, \quad |t| \leqslant \min\{(\Psi(D))^{100}, \ln^3 D\}.$$

Then

$$Q\left(\frac{\Psi(D)}{\ln D}\right) \leqslant \exp(A\Psi(D)), \tag{3}$$

where $A > 0$ is an absolute constant.

Thus, roughly speaking, the quantity of $L$-series with the zeros sufficiently near to $\sigma = 1$ is small. This enables us to treat in a new way the remainder term in the well known formula for primes in a progression.

## § 2. The dissection of the critical stripe

Given a large modulus $D$ we divide the critical stripe $0 \leqslant \sigma \leqslant 1$ into five new stripe:

I. $$0 \leqslant \sigma \leqslant 1 - \frac{(\ln D)^{0,001}}{\ln D}.$$

II. $$1 - \frac{(\ln D)^{0,001}}{\ln D} \leqslant \sigma \leqslant 1 - \frac{\ln \ln D}{\ln D}.$$

III. $$1 - \frac{\ln \ln D}{\ln D} \leqslant \sigma \leqslant 1 - \frac{K_0}{\ln D},$$

$K_0$ being a large constant which will be fixed in the course of the proof.

IV. $$1 - \frac{K_0}{\ln D} \leqslant \sigma \leqslant 1 - \frac{c_0}{\ln D},$$

$c_0$ being a small constant such that in the rectangle $1 \geqslant \sigma \geqslant 1 - c_0/\ln D$, $|t| \leqslant D$, there are no zeros of $L(s, \chi)$'s, except, possibly, one real zero of a series $L(s, \chi)$ belonging to an «exclusive» real character (mod $D$). The existence of such a $c_0$ is proved in [5].

V. $$1 - \frac{c_0}{\ln D} \leqslant \sigma \leqslant 1.$$

We shall characterize these stripes by their properties. The stripe I will be called «the stripe of deep zeros». This stripe even with any fixed $\lambda > 0$ instead of 0,001 was narrowly examined in my paper «On Dirichlet's $L$-series and prime-number sums» [4]. The case of the basic theorem with $\Psi(D) \geqslant (\ln D)^{\lambda}$ (i. e. the estimate (3) with the restriction $\Psi(D) \geqslant (\ln D)^{\lambda}$) was there proved in detail, and the connection with prime-number estimates of acad. I. M. Vinogradow's type was established.

The stripe II will be called «the middle stripe». It requires a more complicated analytical treatment than that of the stripe I in the paper [4], but no new arithmetical theorems.

The stripe III is called «Viggo Brun's stripe». Here the analytical machinery in use is the same as that of the stripe II, but it requires a generalization of that well known Brun—Titchmarsh's theorem [6], which is perhaps the most profound result in the theory of primes in progressions with a variable modulus. The generalization in question was kindly communicated to the author by A. A. Buchstab.

The stripe IV is called «the stripe of the normal density». If $|s-s_0|\leqslant 1/\ln D$ is any circle with the center in the segment $|t|\leqslant D$ of this stripe, then it contains no more than $C(K_0)$ zeros of any separate series $L(s, x)$, where $C(K_0)$ depends on $K_0$ only.

This is a consequence of the «density lemma» proved in my paper [3]; and provides an easy approach to this stripe.

The stripe V is called «Siegel's stripe». In the segment $|t|\leqslant D$ of this stripe can lie only one real zero of one real $L(s, \chi)$ (mod $D$). Hence this stripe may be neglected in the course of the proof of the basic theorem (3). But it is very essential for the proof of the fundamental relations (1) and (2), and requires a careful study of the «Deuring—Heilbronn effect» in the second section of this paper.

## § 3

The stripe I was analysed in detail in [4], and the estimate (3) was proved for $\Psi(D)\geqslant(\ln D)^{0,001}$. Hence we have now to study the series $L(s, \chi)$ with the zeros in the appropriate segments of the stripes II, III and IV.

Our reasonings will form an adaptation of the fundamental scheme given in the paper [1] so as to avoid the use of the hypothetic property (d) which I was unable to prove. The stripe of normal density (the stripe IV) can be treated even without any adaptation.

We introduce now some useful notations.

### Notations

$\mu=\frac{1}{\ln D}$; $L(s, \chi)$ will always denote an $L$-series with a primitive or non-primitive $\chi$ (mod $D$), $L(s, \chi)$ being a determined series.

$(M, Y, z)$-circle means the circle $|s-z|\leqslant Y$ containing $M$ zeros of $L(s, \chi)$.

$(\leqslant M, Y, z)$-circle and $(\geqslant M, Y, z)$-circle are analogous notations.

$\mathfrak{S}[\sigma;\ |t-\eta_0|\leqslant N]$ means the segment of points $\sigma+it$ with $|t-\eta_0|\leqslant N$.

The expression «the segment $\mathfrak{S}_1$ bears the circle $\mathfrak{G}$» means that the center of $\mathfrak{G}$ belongs to $\mathfrak{S}_1$.

$K_0, K_1, K_2, \ldots$ are constants $>1$ each depending only upon the preceding ones, so that, for instance, $K_3=K_3(K_0, K_1, K_2)$.

$C_1, C_2, C_3, \ldots$ is another set of constants with the same property.

The complex variable will be denoted by $s=\sigma+it$, or, sometimes, by $w=\sigma+it$.

The generalized Dirichlet's series of the type

$$\Psi(x) = \sum_{n=-\infty}^{\infty} \Gamma_k \exp[(-\sigma_k + it_k)x]$$

with real $x$, which is absolutely convergent for $x \geqslant 0$, will be called exponential series.

### § 4. Two analytical lemmas

Lemma I. *If $F(s)$ is regular and $\left|\frac{F(s)}{F(s_0)}\right| < e^M$ in the circle $|s-s_0| \leqslant r$, $M > 1$, then*

$$\left|\frac{F'}{F}(s) - \sum_{\rho} \frac{1}{s-\rho}\right| < C_1 \cdot \frac{M}{r},$$

*if $\rho$ runs over the zeros of $F(s)$ in $|s-s_0| \leqslant \frac{r}{2}$.*

Proof. This is the first lemma of Ch. I; 1.4 of the well known book «The zeta-function of Riemann» by E. C. Titchmarsh (Cambr. Tracts, 1930).

The second lemma relates to the exponential series, and is, in essence, a generalization of certain simple properties of almost periodic functions of a special type.

Lemma II. *Let $\Delta$ be a large number, $\Psi_\Delta$ a number under conditions $\Delta^{0{,}001} \geqslant \Psi_\Delta \geqslant 2$.*

*Let $\rho_k = \beta_k + it_k$ ($k = 0, \pm 1, \pm 2, \ldots$) be an infinite sequence of numbers under conditions:*

(1) $\beta_0 \in \left[1 - \frac{\Psi_\Delta}{\Delta}, 1\right]$.

(2) $\beta_k \in (0, 1)$.

(3) $\beta_k \leqslant \beta_0 + \frac{1}{\Delta}$ *for* $|t_0 - t_k| \leqslant \Psi_\Delta^{100}$.

(4) *If $r \in [\Psi_\Delta, 2\Delta]$, then any circle $|s - 1 - ti| \leqslant \frac{r}{\Delta}$ contains no more than $A_1\left(r + \frac{1}{\Delta}\ln(|t|+2)\right)$ $\rho_n$'s, $A_1 > 0$ being a constant.*

(5) *If $z_0$ is any point of the segment $\mathfrak{S}\left[\beta_0 + \frac{2}{\Delta}, |t - t_0| \leqslant \Psi_\Delta^{40}\right]$, and $r \in [1, \Psi_\Delta]$, then the circle $|s - z_0| \leqslant \frac{r}{\Delta}$ contains no more than $A_2(r+1)$ points $\rho_k$.*

*We consider now the exponential series*

$$\Psi_0(x) = \sum_{k=-\infty}^{\infty} \Gamma(\rho_k - it_k) \exp(-\sigma_k + it_k)x,$$

*where $\Gamma$ is Euler's function, and $\tau_1$ any real number such that $|t_0 - \tau_1| \leqslant 1$. We assert that there exist three other constants: $A_5 > A_4 > A_3 > 20$ depending upon $A_2$ and $A_1$ only, such that*

$$\int_{A_3\Delta}^{A_4\Delta} |\Psi_0(x)|^2 \, dx > \frac{\Delta}{A_5}. \tag{4}$$

Proof. The estimate (4) is weaker than the analogous assertion which is proved in detail in my paper «On the characters of primes. I» [3] (§§ 11—17; the relation (13), § 16).

The outline of the proof in a simple particular case is given in [2]. The proof proceeds in using iterated integration.

## § 5

We consider now any fixed $\Psi(D) \in [K_0, (\ln D)^{0{,}001}]$.

Lemma III. *If an $L(s,\chi)$ has a zero in the rectangle $1 \geqslant \sigma \geqslant 1 - \frac{\Psi(D)}{\ln D}$, $|t| \leqslant (\Psi(D))^{100}$ and the constant $K_0$ is sufficiently large, then it has a zero $\rho_0 = \beta_0 + i\eta_0$ under conditions:*

(1) $\beta_0 \geqslant 1 - \frac{\Psi(D)}{\ln D}$, $|\eta_0| \leqslant (\Psi(D))^{102}$.

(2) *There are no other zeros in*

$$\sigma \geqslant \beta_0 + \frac{1}{\ln D}; \quad |t - \eta_0| \leqslant (\Psi(D))^{100}.$$

Proof — by the same «shifting process» as in the proof of the analogous «first lemma» in [4] (§ 3).

Lemma IV. *Any circle with the radius $R \in \left[\frac{1}{\ln D}, 2\right]$ and the center in the point $1 + it$ contains no more than $C_2 R \ln D(|t|+2)$ zeros of $L(s,\chi)$ counted with their multiplicity.*

Proof. This is the trivial generalization of the «density lemma» proved in detail in [3] (§ 8).

## § 6

Lemma V. *If $s = \sigma + it$; $0 \leqslant \sigma \leqslant 2$,*

$$\left| \frac{L'}{L}(s,\chi) - \sum_{|\rho - s| \leqslant 1} \frac{1}{s-\rho} \right| \leqslant C_3 \ln D(|t|+2), \tag{5}$$

*where $\rho$ runs over zeros of $L(s,\chi)$ in the circle $|w-s| \leqslant 1$.*

Proof. Taking into account that on the circle $|w-s| \leqslant 6$ we have:

$$|L(s,\chi)| < \exp(8 \ln D(|t|+2)), \quad |L(s+2)| > \frac{1}{4}$$

(this is the consequence of the well known properties of $L$-series [6]), we deduce our lemma easily from the Lemmas I and IV.

## § 7

In what follows, $L(s,\chi)$ will denote a series which has a zero in $\sigma \geqslant 1 - \frac{\Psi(D)}{\ln D}$, $|t| \leqslant (\Psi(D))^{100}$. We choose $K_0$ to be equal to $(C_2 C_3 + 10)^{1000}$, $K_1 = K_0^{0{,}1}$, $K_2 = K_1^2$.

Let $\rho_0 = \beta_0 + i\eta_0$ be the zero of the Lemma III. We take now $\alpha_0 = \beta_0 + K_1\mu$ and consider the segment

$$\mathfrak{S}_{\alpha_0} = \mathfrak{S}[\alpha_0; |t - \eta_0| \leqslant (\Psi(D))^{70}].$$

If $R \geqslant \Psi(D)$, then, in virtue of the Lemma IV and of the inequality $\beta_0 \geqslant 1 - \frac{\Psi(D)}{\ln D}$, any circle $|s - z_0| \leqslant R\mu$ with $z_0 \in \mathfrak{S}_{\alpha_0}$ contains no more than $10C_2R$ zeros of $L(s,\chi)$, i. e. is a $(\leqslant 10C_2R,\ R\mu,\ z_0)$-circle, according to our terminology (§ 3). We consider now two cases:

I. The analogous property holds also for the circles with $R\in[4K_1, \Psi(D)]$ or more precisely: any circle $|s-z_0|\leqslant R\mu$ with $z_0\in\mathfrak{S}_{\alpha_0}$, $R\in[4K_1, \Psi(D)]$ is a $(\leqslant K_2\cdot R,\ R\mu, z_0)$-circle.

II. The property does not hold, and there is a $z_0\in\mathfrak{S}_{\alpha_0}$ and an $R\in[4K_1, \Psi(D)]$ such that the circle $|s-z_0|\leqslant R\cdot\mu$ is a $(\geqslant K_2R, R\mu, z)$-circle.

## § 8

Lemma VI. *If* $Q_1\left(\frac{\Psi(D)}{\ln D}\right)$ *is the quantity of the series* $L(s,\chi)$ *each having a zero in* $1\geqslant\sigma\geqslant 1-\frac{\Psi(D)}{\ln D}$, $|t|\leqslant(\Psi(D))^{100}$ *and belonging to the case* I *of the preceding paragraph, then*

$$Q_1\left(\frac{\Psi(D)}{\ln D}\right)\leqslant\exp(K_3\Psi(D)). \tag{6}$$

Proof. We develop in detail the scheme of the paper [1]. Let $L(s,\chi)$ be one of our series. Then we shall have (cf. [1], (3)):

$$-\sum_{n=1}^{\infty}\chi(n)\Lambda(n)\,n^{-i\tau_1}e^{-\delta n}=\sum_{\rho_k\in\mathfrak{S}} m_k\Gamma(\rho_k-i\tau_1)\,\delta^{i\tau_1}\delta^{-\rho_k}+O(\ln^2 D). \tag{7}$$

Here $\rho_k$ runs over the zeros of the critical stripe; $\mathfrak{S}$ is the multiplicity of $\rho_k$ and $\tau_1$ is any number under condition $|\tau_1-\eta_0|\leqslant 1$.

Denoting here $(-\ln\delta)=x$ we can write

$$\left|\sum_{\rho_k\in\mathfrak{S}} m_k\Gamma(\rho_k-i\tau_1)\,\delta^{i\tau_1}\delta^{-\rho_k}\right|=$$

$$=\delta^{-\beta_0-\mu}\left|\sum_{k=-\infty}^{\infty}\Gamma(\rho_k-i\tau_1)\exp(-\sigma_k+i\tau_1)x\right|=\exp(\beta_0+\mu)x\cdot|\Psi_0(x)|,$$

where $\rho_k$'s are counted with their multiplicity and $\Psi_0(x)$ is the series of Lemma II, § 5, with $\Delta=\ln D$; $\Psi_\Delta=\Psi(D)$. We denote now

$$\Phi(x,\chi,\tau_1)=\sum_{n=1}^{\infty}\chi(n)\Lambda(n)\,n^{i\tau_1}e^{-\delta n}$$

so that

$$|\Phi(x,\chi,\tau_1)|=\exp(\beta_0+\mu)x\,|\Psi_0(x)|+O(\ln^2 D). \tag{8}$$

We have now by the Lemma II

$$\int_{K_4\cdot\ln D}^{K_5\cdot\ln D}|\Psi_0(x)|^2\,dx\geqslant\frac{\ln D}{K_5}. \tag{9}$$

Hence we obtain, putting $X_1 = K_4 \ln D$, $X_2 = K_5 \ln D$:

$$\int_{X_1}^{X_2} \left| \frac{\Phi(x,\chi,\tau_1)}{e^{(\beta_0+\mu)x}} \right|^2 dx = \int_{X_1}^{X_2} \left| \Psi_0(x) + \frac{\theta \cdot \ln^2 D}{e^{(\beta_0+\mu)x}} \right|^2 dx \geqslant$$

$$\geqslant \int_{X_1}^{X_2} |\Psi_0(x)|^2 dx - 2\ln^2 D \int_{X_1}^{X_2} |\Psi_0(x)|\, e^{-(\beta_0+\mu)x} dx - \ln^4 D \int_{X_1}^{X_2} e^{-2(\beta_0+\mu)x} dx.$$

We have now by the properties of $\Psi_0(x)$:

$$\int_{X_1}^{X_2} |\Psi_0(x)|\, dx < \ln^2 D\, |X_1 - X_2| < \ln^4 D,$$

$$e^{-(\beta+\mu)X_1} < D^{-1/2}$$

and hence

$$\int_{X_1}^{X_2} \left| \frac{\Phi(x,\chi,\tau_1)}{e^{(\beta_0+\mu)x}} \right|^2 dx \geqslant \int_{X_1}^{X_2} |\Psi_0(x)|^2 dx - 2D^{-1/2} \geqslant \frac{\ln D}{2K_6}. \tag{10}$$

Let now $Q_1\left(\frac{\Psi(D)}{\ln D}\right) > \exp(10^6\,\Psi(D))$ and let $L(s,\chi_{\alpha_1})$, $L(s,\chi_{\alpha_2})$, ... ..., $L(s,\chi_{\alpha_{Q_1}})$ be all the corresponding series. For each series we fix the corresponding $\rho_0 = \beta_0 + i\eta_0$ and consider the numbers $\tau_1$ with $|\tau_1 - \eta_0| \leqslant 1$. As $|\eta_0| \leqslant (\Psi(D))^{102}$, by Lemma III, we can choose

$$> \frac{1}{(\Psi(D))^{200}} \cdot Q_1\left(\frac{\Psi(D)}{\ln D}\right)$$

series $L(s,\chi_{\alpha_1}),\ \ldots,\ L(s,\chi_{\alpha_{Q_1}})$ for which we can fix one and the same number $\tau_1$.

Taking into account that $\beta_0 + \mu \geqslant 1 - \frac{\Psi(D)}{\ln D}$ we can only strengthen the inequality (10) replacing $\beta_0+\mu$ by $1 - \frac{\Psi(D)}{\ln D}$ and hence we obtain for $\chi = \chi_{\alpha_1}, \chi_{\alpha_2}, \ldots, \chi_{\alpha_{Q_2}}$:

$$\int_{X_1}^{X_2} |\Phi(x,\chi,\tau_1)|^2 \exp\left(2\frac{\Psi(D)}{\ln D} - 2\right) x\, dx \geqslant \frac{\ln D}{2K_6}, \tag{11}$$

where $\tau_1$, $X_1$, $X_2$ are the same for all $\chi_{\alpha_j}$'s. Hence, on summing the inequalities (11) for our $\chi_{\alpha_j}$'s, we obtain:

$$\int_{X_1}^{X_2} \sum_{\alpha=1}^{Q_2} |\Phi(x,\chi_{\alpha_j},\tau_1)|^2 \cdot \exp\left(\frac{2\Psi(D)}{\ln D} - 2\right) x\, dx \geqslant Q_2\left(\frac{\Psi(D)}{\ln D}\right) \cdot \frac{\ln D}{K_6}. \tag{12}$$

But we have evidently:

$$\sum_{j=1}^{Q_2} |\Phi(x,\chi_{\alpha_j},\tau_1)|^2 \leqslant \sum_{\chi} |\Phi(x,\chi,\tau_1)|^2, \tag{13}$$

where the sum is extended over all $\chi$'s $(\mathrm{mod}\, D)$ including the principal one. The last sum can be easily evaluated:

$$\sum_{\chi} |\Phi(x,\chi,\tau_1)|^2 \leqslant \varphi(D) \sum_{n=1}^{\infty} \Lambda(n)\, e^{-\delta n} \cdot \sum_{\substack{n_1=1,\ldots,\infty \\ n_1 \equiv n (\mathrm{mod} D)}} \Lambda(n_1)\, e^{-\delta n_1}. \tag{14}$$

Here we apply the crucial

Lemma VII. *The number of prime numbers in a segment of progression* $Dx+l, l\in(0,D)$, $1\leqslant Dx+l\leqslant N$ *does not surpass* $C_4\cdot\frac{N}{\varphi(D)\ln N}$.

Proof. This is a particular case of Brun—Titchmarsh's theorem [6]. Hence, if $\delta^{-1}>D^4$, we deduce:

$$\sum_{n=1}^{\infty}\Lambda(n)\,e^{-\delta n}<C_5\delta^{-1},$$

$$\sum_{n_1\equiv n\,(\mathrm{mod}\,D)}\Lambda(n_1)\,e^{-\delta n_1}<C_5\cdot\frac{\delta^{-1}}{\varphi(D)}.$$

And so, for $\delta<D^{-4}$, or $x>4\ln D$,

$$\sum_{\chi}|\Phi(x,\chi,\tau_1)|^2<C_6\delta^{-2}=C_6\cdot e^{2x}.$$

Hence, returning to (12), we obtain: as $X_1>4\ln D$—

$$\int_{X_1}^{X_2}C_6e^{2x}\cdot e^{2\frac{\Psi(D)}{\ln D}x-2x}dx\geqslant Q_2\left(\frac{\Psi(D)}{\ln D}\right)\cdot\frac{\ln D}{K_6}$$

or

$$\int_{X_1}^{X_2}e^{2\frac{\Psi(D)}{\ln D}x}\,dx\geqslant\frac{1}{K_7}Q_2\left(\frac{\Psi(D)}{\ln D}\right)\cdot\ln D.$$

Calculating the integral on the left hand side, we get:

$$\frac{\ln D}{\Psi(D)}\cdot\exp\left(2X_2\cdot\frac{\Psi(D)}{\ln D}\right)\geqslant\frac{\ln D}{K_7}\cdot Q_2\left(\frac{\Psi(D)}{\ln D}\right)$$

or, as $X_2=K_5\ln D_1$,

$$Q_2\left(\frac{\Psi(D)}{\ln D}\right)\leqslant\exp\left(K_8\Psi(D)\right).$$

As $Q_2\left(\frac{\Psi(D)}{\ln D}\right)\geqslant\left\{Q_1\left(\frac{\Psi(D)}{\ln D}\right)\right\}^{1/2}$, we have

$$Q_1\left(\frac{\Psi(D)}{\ln D}\right)\leqslant\exp\left(K_3\Psi(D)\right)$$

which proves Lemma VI.

## § 9

On ground of this fact we can consider now only the case II of § 8 in the course of the proof of the basic theorem. In what follows, $L(s,\chi)$ will be the series which has a zero in $1\geqslant\sigma\geqslant1-\frac{\Psi(D)}{\ln D}$, $|t|\leqslant(\Psi(D))^{102}$ and hence a zero $\rho_0=\beta_0+i\eta_0$ of the Lemma III. The series $L(s,\chi)$ will be supposed to belong to the case II of § 8.

Hence there exists a $(\geqslant K_2R,\ R\mu,\ s_0)$-circle with $s_0\in\mathfrak{S}(\alpha_0;\ |t-\eta_0|\leqslant\leqslant(\Psi(D))^{70})$, $\alpha_0=\beta_0+K_1\mu$, $K_2=K_1^2$, $R\in[4K_1,\ \Psi(D)]$.

Lemma VIII. *$L(s,\chi)$ being the series considered above, there exists a point $z_1=\alpha_1+i\eta_1$ under following conditions:*

(1) *$\alpha_1=\beta_0+2^{j_1}K_1\cdot\mu=\beta_0+\Delta_1\mu$ with $j_1\in[1, 2\ln\Psi(D)]$ integer; $|\eta_1|\leqslant(\Psi(D))^{75}$.*

(2) *The circle $|s-z_1|\leqslant 4\cdot 2^{j_1}\cdot K_1\mu=4\Delta_1\mu$ is a $(P\Delta_1, 4\Delta_1\mu, z_1)$-circle with*

$$P\in\left[\frac{K_2}{10},\ 2C_2\Psi(D)\right].$$

(3) *The segment $\mathfrak{S}[\alpha_1; |t-\eta_1|\leqslant(\Psi(D))^{50}]$ bears only $(\leqslant 2P\Delta_1, 4\Delta_1\mu, z)$-circles.*

(4) *The segments $\mathfrak{S}[\alpha_1; |t-\eta_1|\leqslant(\Psi(D))^{50}]$ with $j=1,2,\ldots,[2\ln\Psi(D)]$, $\sigma_j=\alpha_1+2^j\Delta_1\mu$ bear only $(\leqslant 8K_2\cdot 2^j\Delta_1, 4\cdot 2^j\Delta_1\mu, z)$-circles.*

Proof. We return to our $(\geqslant K_2R, R\mu, s_0)$-circle. Let $R\in[2^lK_1, 2^{l+1}K_1]$ and let $s_0'=\beta_0+2^lK_1\mu+iIs_0$ be the center of a $(\geqslant K_2\cdot 2^lK_1, 4\cdot 2^l\cdot K_1\mu, s_0')$-circle. If the segments $\mathfrak{S}[\beta_0+2^mK_1\mu, |t-Is_0|\leqslant(\Psi(D))^{60}]$ for $m>l$ bear only $(\leqslant 8K_2\cdot 2^mK_1, 4\cdot 2^mK_1\mu, z)$-circles, our first process is finished. If this is not so, we take the $(\geqslant 8K_2\cdot 2^mK_1, 4\cdot 2^mK_1\mu, s_0'')$-circle with the greatest possible $m$ (evidently, $m\leqslant 2\ln\Psi(D)$ in virtue of the Lemma IV), $s_0''$ belonging to one of the segments of our set, so that $|Is_0''-Is_0|\leqslant(\Psi(D))^{60}$. We have: $|Is_0''|\leqslant 2(\Psi(D))^{70}$. We repeat our process beginning with the $(\geqslant 8K_2\cdot K_1\cdot 2^m, 4\cdot K_12^m\cdot\mu, s_0'')$-circle and so, taking account of the Lemma IV, we shall find in no more than $2\ln\Psi(D)$ steps a point $z_0'$ with $|Iz_0'|<(\Psi(D))^{72}$ which is the center of a $(\geqslant 8K_2\cdot 2^n\cdot K_1, 4\cdot 2^n\cdot K_1\mu, z_0')$-circle wheras all the segments $\mathfrak{S}(\beta_0+2^q\cdot K_1\mu, |t-Iz_0'|\leqslant(\Psi(D))^{60}$ bear only $(\leqslant 8K_2\cdot 2^q\cdot K_1\mu, 4\cdot 2^q\cdot K_1\mu, z)$-circles, for $q>n$.

We denote now our circle with the center $z_0'$, by $(P_0'\cdot 2^{j_1}\cdot K_1, 4\cdot 2^{j_1}\cdot K_1\mu, z_0')$-circle; here $n=j_1$ and $P_0'\geqslant 8K_2$; we put moreover $\alpha_1=\beta_0+2^{j_1}\cdot K_1\mu$. If the segment $\mathfrak{S}[\alpha_1, |t-Iz_0'|\leqslant(\Psi(D))^{55}]$ bears a $(\geqslant 2P_0'\cdot 2^{j_1}K_1, 4\cdot 2^{j_1}\cdot K_1\mu, z_0'')$-circle, we put $P_0''$ equal to the quantity of zeros of $L(s,\chi)$ in this circle, and consider the segment $\mathfrak{S}[\alpha_1, |t-Iz_0''|\leqslant(\Psi(D))^{55}]$. The process will come to the end in no more than $(\ln\Psi(D))^2$ steps in virtue of the Lemma IV. This completes our proof.

## § 10

We denote $f(s)=\dfrac{L'}{L}(s,\chi)$, it is regular in $\sigma\geqslant\beta_0+\dfrac{3}{2}\mu$; $|t-\eta_1|\leqslant(\Psi(D))^{80}$.

Lemma IX. *If $\sigma\geqslant\beta_0+2\mu$, $|t-\eta_1|\leqslant(\Psi(D))^{80}$ we have*

$$|f(s)|\leqslant C_7(\Psi(D))^4\ln D. \tag{15}$$

Proof. For $\sigma>1+\mu$, $|f(s)|<C_8\ln D$ (cf. [6]). Let $s=x+iy$ with $\beta_0+2\mu\leqslant x\leqslant 1+\mu$, $|\eta_1-y|\leqslant(\Psi(D))^{80}$. We have by Lemma V:

$$\left|f(s)-\sum_{|\rho-\bar{s}|\leqslant 1}\frac{1}{s-\rho}\right|\leqslant 2C_3\ln D$$

and so

$$f(s)=\sum_{|\rho-s|\leqslant 1}\frac{1}{s-\rho}+\theta\cdot 2C_3\ln D,\quad |\theta|\leqslant 1. \tag{16}$$

We put now $s_1=s+1-x+\mu=1+\mu+iy$; hence

$$f(s_1)=\sum_{|\rho-s_1|\leqslant 1}\frac{1}{s_1-\rho}+\theta\cdot 2C_3\ln D.$$

But the Lemma IV implies:

$$\sum_{|s_1-\rho|\leqslant 1}\frac{1}{s_1-\rho}=\sum_{|s-\rho|\leqslant 1}\frac{1}{s_1-\rho}+\theta\cdot C_9\ln D.$$

Hence

$$f(s)-f(s_1)=\sum_{|\rho-s|\leqslant 1}\frac{s_1-s}{(s_1-\rho)(s-\rho)}+\theta\cdot C_{10}\ln D. \tag{17}$$

We draw now the circles:

$$|w-s|\leqslant(\Psi(D))^2\cdot\mu=R\mu,\ |w-s|\leqslant 2R\mu,\ \leqslant 2^2R\mu,\ \dots,\ |w-s|\leqslant 2^q\cdot R\mu,$$

$$2^qR\leqslant\ln D,\ 2^{q+1}R>\ln D.$$

By Lemma IV, the quantity of $\rho$'s in such a circle, $|w-s|\leqslant 2^jR\mu$ is $<C_2\cdot\Psi(D)\cdot 2^jR=C_2\cdot 2^j(\Psi(D))^3$. If $\rho$ belongs to the circle $|w-s|\leqslant R\mu$, we have

$$\frac{|s-s_1|}{|(s-\rho)(s_1-\rho)|}<2(\Psi(D))^2\cdot\mu\cdot\ln^2 D<2(\Psi(D))^2\ln D. \tag{18}$$

The quantity of such $\rho$'s, is $<2C_2(\Psi(D))^2$. If $\rho$ belongs to $2^{j-1}R\mu\leqslant|w-s|\leqslant 2^jR\mu$, then $|s-\rho|>\frac{1}{2}\cdot 2^jR\mu$; $|s_1-\rho|>\frac{1}{2}\cdot 2^j\cdot R\mu$ and hence the corresponding sum does not surpass

$$2(\Psi(D))^2\cdot\mu\cdot\frac{4}{2^{2j}R^2\mu^2}\cdot 2C_2\Psi(D)\cdot 2^jR<\frac{16}{R}\cdot\frac{1}{2^j}(\Psi(D))^3\ln D.$$

Summing up these estimates for $j=1,2,\dots,q$ and taking into account (18), we get

$$\left|\sum_{|s-\rho|\leqslant 1}\frac{s_1-s}{(s-\rho)(s_1-\rho)}\right|\leqslant C_{10}(\Psi(D))^4\cdot\ln D.$$

Returning to (17) we get

$$|f(s)-f(s_1)|\leqslant C_{11}(\Psi(D))^4\cdot\ln D.$$

As $f(s_1)<C_8\ln D$ the lemma is proved.

### § 11. The auxiliary function $B_\Psi(s)$

We introduce now the number analogous to that of the paper [4]:

$$N_\Psi=\exp\left(0{,}001\frac{\ln D}{\Psi(D)}\ln\Psi(D)\right) \tag{19}$$

and the function

$$B_{\Psi}(s) = -\sum_{p \leqslant N_{\Psi}} \frac{\chi(p)\ln p}{p^s} - \sum_{m=2}^{\infty}\sum_{p} \frac{\chi(p)\ln p}{p^{ms}}. \tag{20}$$

Here $p$ denotes the typical prime. We have obviously in $\sigma > 1$:

$$f(s) - B_{\Psi}(s) = \frac{L'}{L}(s, \chi) - B_{\Psi}(s) = -\sum_{p \leqslant N_{\Psi}} \frac{\chi(p)\ln p}{p^s}.$$

For $\sigma > \frac{5}{9}$ we have obviously

$$B_{\Psi}(s) = -\sum_{p \leqslant N_{\Psi}} \frac{\chi(p)\ln p}{p^s} + O(1).$$

Lemma X. *In* $\sigma \geqslant 1 - \frac{R}{\ln D}$, $10\Psi(D) \leqslant R \leqslant \frac{4}{9}\ln D$ *we have*

$$|B_{\Psi}(s)| < C_{12}\frac{\ln D}{\Psi(D)} \cdot \exp\left(0{,}001\, \frac{R}{\Psi(D)} \ln \Psi(D)\right). \tag{21}$$

*In particular, in* $\sigma \geqslant 1 - \frac{10\,\Psi(D)}{\ln D}$,

$$|B_{\Psi}(s)| \leqslant \frac{\ln D}{(\Psi(D))^{2/3}}. \tag{22}$$

Proof. We have in $\sigma \geqslant 1 - \frac{R}{\ln D}$ by Tshebycheff's estimates:

$$|B_{\Psi}(s)| \leqslant \sum_{n \leqslant N_{\Psi}} \frac{\Lambda(n)}{n^{1-R/\ln D}} + O(1) \leqslant 100 \sum_{n \leqslant N_{\Psi}} \frac{1}{n^{1-R/\ln D}} \leqslant$$

$$\leqslant 100 N_{\Psi}^{\frac{R}{\ln D}} \cdot \sum_{n \leqslant N_{\Psi}} \frac{1}{n} \leqslant 200 \ln N_{\Psi} \cdot N_{\Psi}^{\frac{R}{\ln D}} =$$

$$= C_{12}\frac{\ln D}{\Psi(D)} \cdot \ln \Psi(D) \cdot \exp\left(0{,}001\, \frac{R}{\ln D} \ln \Psi(D)\right).$$

This is (21). For $R = 10\,\Psi(D)$, we get:

$$\exp\left(0{,}001\, \frac{R}{\Psi(D)} \cdot \ln \Psi(D)\right) = \exp(0{,}01 \ln \Psi(D)) = (\Psi(D))^{0{,}01};$$

hence

$$|B_{\Psi}(s)| \leqslant 200\frac{\ln D}{(\Psi(D))^{0{,}99}}.$$

As $\Psi(D) > K_0$ and our $K_0$ is $> 10^{1000}$ we obtain (22).

Returning now to the Lemma VIII, we take the number $\eta_1$ and fix an arbitrary real number $\tau_0$ such that $|\tau_0 - \eta_1| \leqslant 1$.

We introduce now the function

$$f_1(s) = f(s) - B_{\Psi}(s) = \frac{L'}{L}(s, \chi) - B_{\Psi}(s) \tag{23}$$

and form the basic function

$$\varphi(s) = \frac{f_1(s)}{s - i\tau_0}. \tag{24}$$

This function is regular in $\sigma \geqslant 1$ and in the rectangle $\beta_0 + 2\mu \leqslant \sigma \leqslant 1$, $|t - \eta_1| \leqslant (\Psi(D))^{80}$.

## § 12

The leading idea of the following arguments is to replace the classical function $\frac{L'}{L}(s, \chi)=f(s)$ by a new formation $Q(w, Z)=\frac{\varphi'(w)}{\varphi(w)-Z}$, where $Z$ is fixed so as to facilitate the study of the distribution of its «principal» poles, and then, utilizing the obtained results, to deduce some facts relating to $\frac{L'}{L}(s, \chi)$ which enable to prove (3).

I was led to make such a detour after many unsuccessful attempts to prove the «density property» (d).

## § 13. The maxima of $|\varphi(s)|$ on the vertical segments

We proceed now to the choice of a convenient value of $Z$. It proves to be advantegeous to take it as a maximal value of $|\varphi(s)|$ on a certain segment, multiplied by $e^{i\vartheta}$ with a suitable $\vartheta$.

We consider the set of all the rectangles with the sides:

$$\sigma=\sigma' \geqslant \beta_0+2\mu, \quad \sigma=2, \quad t-\tau_0=(\Psi(D))^{80}, \quad t-\tau_0=-(\Psi(D))^{80}.$$

The function $\varphi(s)$ is regular on these rectangles including the contour, and, by (24), (23), (22) and (15) (Lemma IX), we have

$$|\varphi(s)|<\frac{C_7(\Psi(D))^4 \ln D}{|s-i\tau_0|}. \tag{25}$$

We assert now that the maximum of $|\varphi(s)|$ on the rectangle is either attained on the left vertical side

$$\sigma=\sigma', \text{ or } |\varphi(s)|<C_7\frac{\ln D}{(\Psi(D))^{75}}.$$

In fact, by Cauchy's principle this maximum is attained on the contour; on $\sigma=2$, $|\varphi(s)|=O(1)$, and on horizontal sides

$$|\varphi(s)|<C_7\frac{\ln D}{(\Psi(D))^{75}}.$$

Lemma XI. *On $\mathfrak{S}[\alpha_1, |t-\tau_0|\leqslant(\Psi(D))^{80}]$, where $\alpha_1$ is the number introduced in Lemma VIII, we have:*

$$\sup|\varphi(s)|=P_0 \ln D>\frac{P \ln D}{1000}. \tag{26}$$

Proof. We compute $|f(z_1)|$ (Lemma VIII). We have by (5) (Lemma V)

$$\Re f(z_1)=\sum_{|z_1-\rho|\leqslant 1}\frac{1}{z_1-\rho}+\theta\cdot 2C_3 \ln D.$$

As we have for all such $\rho$'s, $\Re\frac{1}{z_1-\rho}<0$, we can write

$$|\Re f(z_1)|\geqslant\sum_{|z_1-\rho|\leqslant 4\Delta_1\mu}\left|\Re\frac{1}{z_1-\rho}\right|+\theta\cdot 2C_3 \ln D.$$

We have obviously $\left|\Re\frac{1}{z_1-\rho}\right|>0{,}01\frac{\ln D}{\Delta_1}$, and the quantity of our $\rho$'s is $P\Delta_1$ by Lemma VIII, whence

$$|\Re f(z_1)|>0{,}01\, P\cdot\ln D+\theta\cdot 2C_3 \ln D>0{,}001\cdot P \ln D.$$

In view of (23) and (22), we have thus proved (26).

§ 14

Denoting $M_\sigma$ the maximum of $|\varphi(s)|$ on $\mathfrak{S}[\sigma, |t-\tau_0| \leqslant (\Psi(D))^{80}]$, we see by § 14 that $M_\sigma$ is either $\leqslant C_7 \frac{\ln D}{(\Psi(D))^{75}}$ or is a decreasing function of $\sigma$. In fact, it is obviously non-increasing, and, as $\varphi(s)$ cannot be a constant, it must be strictly decreasing.

Let $\bar{\sigma}$ be the number such that $M_\sigma > C_7 \frac{\ln D}{(\Psi(D))^{75}}$ for $\sigma < \bar{\sigma}$ and $M_\sigma \leqslant C_7 \frac{\ln D}{(\Psi(D))^{75}}$ for $\sigma > \bar{\sigma}$. We have $\bar{\sigma} > \alpha_1$.

We consider now the maxima $M_j = M_{\sigma_j}$ of $|\varphi(s)|$ on the segments

$$\mathfrak{S}_j = \mathfrak{S}[\sigma_j;\ |t-\tau_0| \leqslant (\Psi(D))^{80}], \quad \sigma_j = \beta_0 + 2^j \Delta_1 \mu$$

$j \leqslant l$, $l-1$ being the last integer for which $\sigma_j \leqslant \bar{\sigma}$.

Lemma XII. *There exists an integer $j_0 \in [0, l-1]$ such that*

$$M_{j_0+1} \leqslant \left(1 - \frac{1}{(j_0+2)^2}\right) M_{j_0} \tag{27}$$

*and*

$$M_{j+1} \geqslant \left(1 - \frac{1}{(j+2)^2}\right) M_j \text{ for } j < j_0. \tag{28}$$

Proof. Suppose that $M_{j+1} > \left(1 - \frac{1}{(j+2)^2}\right) M_j$ $(j = 0, 1, \ldots, l-1)$. We have then

$$M_l > \left(1 - \frac{1}{(l+1)^2}\right)\left(1 - \frac{1}{l^2}\right) \cdots \left(1 - \frac{1}{2^2}\right) M_0 > \frac{M_0}{1000},$$

$$M_0 < 1000 M_l \leqslant 1000 C_7 \frac{\ln D}{(\Psi(D))^{75}}.$$

But $M_0 = P_0 \ln D > \frac{P \ln D}{1000}$ by (26), which is impossible. Hence the lemma is proved.

§ 15

We consider now two different cases:

I. $j_0 = 0$, $M_1 \leqslant \left(1 - \frac{1}{2^2}\right) M_0$.

II. $j_0 > 0$, $M_{j_0+1} \leqslant \left(1 - \frac{1}{(j_0+2)^2}\right) M_j$.

Let $M_{j_0}$ be the maximum in question; by what precedes, $M_{j_0} > 0{,}001 M_0$ and so, $M_{j_0}$ is attained on the segment $\mathfrak{S}[\sigma_j;\ |t-\tau_0| \leqslant (\Psi(D))^{10}]$ (cf. (25)).

Let $\nu$ be the point of this segment which is the nearest to $i\tau_0$ such that $|\varphi(\nu)| = M_{j_0}$.

Lemma XIII. *If $z$ belongs to $\mathfrak{S}[\sigma_{j_0};\ |t-\tau_0| \leqslant (\Psi(D))^{45}]$, then the equation $\varphi(s) - \varphi(\nu) = \varphi_1(s) = 0$ has in the circle $|s-z| \leqslant 10^{-5} \cdot 2^{j_0} \cdot \Delta_1 \cdot \mu$ no more than $K_8 \ln(j_0+2)$ zeros.*

Proof. We have analyse separately the two cases.

Case I. $j_0=0$; $M_1 \leqslant \left(1-\frac{1}{2^3}\right) M_0$. Let $z$ be any point of $\mathfrak{S}[\sigma_0;\ |t-\tau_0| \leqslant (\Psi(D))^{45}]$. We consider firstly the circle $|s-z| \leqslant 10^{-4} \cdot \Delta_1 \cdot \mu$ and we shall prove that in this circle $\varphi(s)| \leqslant K_9 M_0$, $K_9$ depending on $K_2$ only. To this end it is sufficient to prove that $|\varphi(w)| \leqslant K_9 M_0$, $w$ being a point of its circumference.

We have

$$\varphi'(w) = \frac{f_1'(w)}{w-i\tau_0} - \frac{f_1(w)}{(w-i\tau_0)^2}.$$

As the whole circle is situated in $\sigma > \beta_0 + 2\mu$; $|t-\tau_0| \leqslant 2\Psi(D))^5$ we have by (23), (22)

$$\left|\frac{f_1(w)}{(w-i\tau)^2}\right| < 2C_7 (\Psi(D))^4 \ln D < 2(\Psi(D))^4 \cdot M_0,$$

$$|B_\psi'(w)| = \frac{1}{2\pi}\left|\oint\limits_{|w_1-w|=\Delta_1\mu} \frac{B_\psi(w_1)}{(w_1-w)^2} dw_1\right| < \frac{\ln D}{(\Psi(D))^{2/3}\Delta_1\mu} < \frac{M_0}{\Delta_1\mu},$$

$$|f'(w)| = \left|\left(\frac{L'}{L}(w)\right)'\right| = \left|\sum_{\rho_n \in \mathfrak{S}} \frac{1}{(w-\rho_n)^2}\right| + \theta\sqrt{\ln D}. \tag{29}$$

This follows from the decomposition of $\frac{L'}{L}(w)$ into fractions [7].

To evaluate it, we draw the circles

$|w_1 - w| = 8\Delta_1\mu,\ 8\cdot 2\cdot\Delta_1\mu,\ 8\cdot 2^2\cdot\Delta_1\mu,\ \ldots,\ 8\cdot 2^l\cdot\Delta_1\mu \quad (l \leqslant 2\ln\ln D).$

By Lemmas VIII and IV they are all

$$(\leqslant 100K_2 2^n \cdot \Delta_1,\ 8\cdot 2^n \cdot \Delta_1\mu,\ w)\text{-circles}.$$

Hence the sum $\sum \frac{1}{(w-\rho_n)^2}$ with $\rho_n$ situated within these circles, is

$$\leqslant \frac{10^4}{\Delta_1^2\mu^2} \cdot \sum_{j=1}^{2\ln\ln D} K_2 \cdot 2^n \cdot \Delta_1 \cdot \frac{1}{8^2 \cdot 2^{2n}} = \frac{10^4}{64}\frac{K^2}{\Delta_1\mu^2} \cdot \sum_{j=1}^{2\ln\ln D} \frac{1}{2^n} < \frac{1000K_2M_0}{\Delta_1\cdot\mu}.$$

For the remaining zeros $\rho_k$ of the critical stripe we have $\sum \frac{1}{|w-\rho_k|^2} < C_{13}\ln D$.

Summing up these results, we obtain

$$|\varphi'(w)| < \frac{2000\,K_2M_0}{\Delta_1\cdot\mu}.$$

The function $\varphi'(w)$ being regular, the same estimation holds inside the circle and therefore

$$|\varphi(w)-\varphi(z)| < \frac{2000K_2M_0}{\Delta_1\cdot\mu} \cdot 10^{-4}\Delta_1\mu < \frac{K_2}{2}M_0.$$

As $|\varphi(z)| \leqslant |\varphi(v)| = M_0$, we have $|\varphi(w)| < K_2M_0$, q. e. d.

We take now two new points

$$z' = z + \Delta_1\mu \cdot 10^{-6}, \quad z'' = z + \frac{3}{2}\Delta_1\mu$$

and draw three circles:

$$\mathfrak{G}_1:\quad |s-z''|=\frac{3}{2}\Delta_1\mu;\qquad \mathfrak{G}_2:\quad |s-z''|\leqslant\left(\frac{3}{2}-10^{-6}\right)\Delta_1\mu;$$

$$\mathfrak{G}_3:\quad |s-z''|\leqslant\frac{\Delta_1\mu}{2}.$$

On the outer circle $\mathfrak{G}_1$ we have $|\varphi(s)|\leqslant M_0$ by construction, on $\mathfrak{G}_3$ we have $|\varphi(s)|\leqslant\left(1-\frac{1}{2^2}\right)M_0$. Hence, by Hadamard's three circles theorem [8] we get on

$$\mathfrak{G}_1:\quad |\varphi(s)|<\left(1-\frac{1}{C_{14}}\right)M_0;$$

in particular, $|\varphi(z')|<\left(1-\frac{1}{C_{14}}\right)M_0$, $C_{14}>1$ being a suitable constant. As $|\varphi(v)|=M_0$, we have: $|\varphi(z')-\varphi(v)|>\frac{M_0}{C_{15}}$. Hence we shall be able to estimate the number of zeros of $\varphi_1(s)=\varphi(s)-\varphi(v)$ in the circle $|s-z'|<\frac{10^{-4}}{6}\Delta_1\mu$. If $Q$ is this quantity we have by Jensen's theorem [9], making use of preceding evaluations:

$$2^Q\leqslant\frac{2K_2M_0}{|\varphi_1(z')|}<\frac{2K_2M_0}{M_0/C_{14}}=2C_{14}K_2\quad (Q\leqslant K_8\ln 2).$$

As our circle includes the circle $|s-z|\leqslant 10^{-5}\Delta_1\mu$ the lemma is proved in this case.

Case II. $j_0>0$; $M_{j_0+1}\leqslant\left(1-\frac{1}{(j_0+2)^2}\right)M_{j_0}$. This case is easier to study.

Let $|\varphi(v)|=M_{j_0}$ and $z\in\mathfrak{S}[\sigma_{j_0};\ |t-\tau_0|\leqslant(\Psi(D))^{45}]$. We consider the circle $|s-z|\leqslant 2^{j_0}\cdot 10^{-4}\Delta_1\mu$. If $w$ is any point of its circumference, we have by construction $|\varphi(w)|\leqslant M_{j_0-1}$, but $M_{j_0}\geqslant\left(1-\frac{1}{(j_0+2)^2}\right)M_{j_0-1}$, $M_{j_0-1}\leqslant 2M_{j_0}$, $|\varphi(w)|\leqslant 2M_{j_0}$. Now drawing the three circles as in case I with $\Delta_1\mu\cdot 2^{j_0}$ instead of $\Delta_1\mu$, we get by Hadamard's three circles theorem:

$$|\varphi(z+10^{-6}\cdot 2^{j_0}\cdot\Delta_1\mu)\leqslant M_{j_0}\left(1-\frac{1}{(j_0+2)^2}\right)^{1/C_{15}}.$$

Hence

$$|\varphi(z+10^{-1}\cdot 2^{j_0}\Delta_1\mu)-\varphi(v)|>\frac{1}{C_{16}}\frac{M_{j_0}}{(j_0+2)^2}.$$

So that in the circle $|s-(z+10^{-6}\cdot 2^{j_0}\cdot\Delta_1\mu)|\leqslant\frac{10^{-4}\cdot 2^{j_0}\cdot\Delta_1\mu}{6}$ we have the following estimate for the number of zeros $Q$: $2^Q<\frac{3M_{j_0}}{M_{j_0}}C_{16}(j_0+2)^2$ whence $Q<K_8\ln(j_0+2)$ and the lemma is proved.

## § 16

We have

$$|\varphi(v)|=M_{j_0}>\frac{M_0}{1000}>10^{-6}\cdot P\cdot\ln D.$$

We put $\varphi(\nu)=Z$, $|\varphi(\nu)|=|Z|>10^{-6}P\ln D$. We put $r_1=10^{-5}\cdot 2^{j_0}\cdot\Delta_1$. We have seen that $\varphi_1(s)=\varphi(s)-Z$ has no more than $K_8\ln(j_0+2)$ zeros in any circle with the radius $r_1\cdot\mu$ and the center on $\mathfrak{S}[\sigma_{j_0};\ |t-\tau_0|\leqslant(\Psi(D))^{45}]$.

As $K_8\ln(j_0+2)<K_8\ln\ln r_1$, we can say that in each circle of the described type, the number of zeros of $\varphi_1(s)$

$$Q\leqslant K_8\ln\ln r_1. \tag{30}$$

The function $\varphi_1(s)$ is regular in $\sigma\geqslant\beta_0+2\mu$, $|t-\tau_0|\leqslant(\Psi(D))^{70}$, and meromorphic in the whole plane. Its poles coincide with those of $f(s)=\frac{L'}{L}(s,\ \chi)$.

Let $\nu_1$ be any number of the segment $\mathfrak{S}[\sigma_{j_0};\ |t-\tau_0|\leqslant(\Psi(D))^{42}]$ and $R\in[4\Delta_1\cdot 2^{j_0},\ 4\Delta_1\cdot 2^{j_0}\ln\ln D]$. We need now the estimate of the number of zeros of the meromorphic function $\varphi_1(s)$ in the circle.

Lemma XIV. *The number of zeros of $\varphi_1(s)$ in $|s-\nu_1|\leqslant R\mu$ does not exceed $K_{10}R$, $K_{10}$ depending upon $K_1$ only*; $R\leqslant\sqrt{\ln D}$.

Proof. We take the point $\nu_2=\nu_1+\frac{R}{4}\mu$ and draw the three circles:

$\mathfrak{G}_1$: $|w-\nu_2|\leqslant 8R\mu$; $\mathfrak{G}_2$: $|w-\nu_2|\leqslant 16R\mu$; $\mathfrak{G}_3$: $|w-\nu_2|\leqslant 32R\mu$.

We denote their circumferences by $\mathfrak{C}_1$, $\mathfrak{C}_2$, $\mathfrak{C}_3$; the ring between $\mathfrak{G}_1$ and $\mathfrak{G}_3$ by $\mathfrak{B}_{13}$ and the ring between $\mathfrak{G}_3$ and $|s-\nu_1|=1$ by $\mathfrak{B}_4$.

Let $\rho_1,\ \rho_2,\ \ldots,\ \rho_{Q_1}$ be all the zeros $L(w,\ \chi)$ in $\mathfrak{G}_3$. We form the products

$$\prod(w)=\prod_{k=1}^{Q_1}(w-\rho_k)=\prod_{\rho_k\in\mathfrak{G}_3}(w-\rho_k),$$

and

$$G(w)=\varphi_1(w)\cdot\Pi(w)=[\varphi(w)-\varphi(\nu)]\,\Pi(w).$$

This function is regular in $\mathfrak{G}_3$. We need its estimate on $\mathfrak{C}_2$. We have by (23), (16) and Lemma IV:

$$\varphi(w)=\frac{1}{w-i\tau_0}\Big[\sum_{|\rho_k-\nu_1|\leqslant 1}\frac{1}{w-\rho_k}-B_\Psi(w)+\theta\cdot 4C_3\ln D\Big] \tag{31}$$

for $w\in\mathfrak{C}_2$;

$$\varphi(\nu_2)=\frac{1}{\nu_2-i\tau_0}\Big[\sum_{|\rho_k-\nu_1|\leqslant 1}\frac{1}{\nu_2-\rho_k}-B_\Psi(\nu_2)+\theta\cdot 4C_3\ln D\Big]. \tag{32}$$

We introduce the functions

$$\varphi_2(s)=\frac{1}{s-i\tau_0}\sum_{\rho_k\in\mathfrak{G}_1}\frac{1}{s-\rho_k},\quad \varphi_3(s)=\frac{1}{s-i\tau_0}\sum_{\rho_k\in\mathfrak{G}_4}\frac{1}{s-\rho_k}.$$

We need the estimates of $\varphi_2(w)-\varphi_2(\nu_2)$ and $\varphi_3(w)-\varphi_3(\nu_2)$. We have obviously by Lemma VIII that $\mathfrak{G}_1$ contains no more than $1000\,K_2R$ $\rho_k$'s, and therefore

$$\left.\begin{aligned}&|\varphi_2(w)| < 1000K_2R\cdot\frac{1}{R\mu} < |Z|\cdot K_2,\\&|\varphi_2(v_2)| < |Z|\cdot K_2,\\&|\varphi_3(w)-\varphi_3(v_2)| = |\varphi_3'(\xi)|\cdot|w-v_2|,\end{aligned}\right\} \tag{33}$$

where $\xi$ is some point on the segment between $w$ and $v_2$.

$$\varphi_3'(\xi) = -\frac{1}{(\xi-i\tau_0)^2}\sum_{\rho_k\in\mathfrak{B}_4}\frac{1}{\xi-\rho_k}+\frac{1}{\xi-i\tau_0}\sum_{\rho_k\in\mathfrak{B}_4}\frac{1}{(\xi-\rho_k)^2}.$$

Making use of Lemmas VIII and IV we get:

$$\left|\sum_{\rho_k\in\mathfrak{B}_4}\frac{1}{\xi-\rho_k}\right| < \sum_{j=1}^{2\ln\ln D}\frac{1000K_2\cdot 2^j\cdot R}{2^j\cdot R\cdot\mu}\leqslant 2\cdot 10^3K_2\ln D\ln\ln D, \tag{34}$$

$$\left|\sum_{\rho_k\in\mathfrak{B}_4}\frac{1}{(\xi-\rho_k)^2}\right| \leqslant \sum_{j=1}^{2\ln\ln D}\frac{1000K_2\cdot 2^j\cdot R}{2^{2j}R^2\cdot\mu^2} < 4\cdot 10^3K_2\frac{\ln^2 D}{R}. \tag{35}$$

Hence

$$|\varphi_3(w)-\varphi_3(v_2)| < 20R\mu\cdot 4\cdot 10^3\cdot K_2\left(\ln D\cdot\ln\ln D+\frac{\ln^2 D}{R}\right) < 10^6K_2|Z|$$

since $R\ln\ln D < |Z|$, as $R < \sqrt{\ln D}$

$$|\varphi_3(w)-\varphi_3(v_2)| \leqslant 10^6\cdot K_2\cdot|Z|. \tag{36}$$

We have further

$$\left|\sum_{\rho_k\in\mathfrak{B}_{13}}\frac{1}{v_2-\rho_k}\right| < \frac{8K_2\cdot 32R}{8R\mu} < 32|Z|. \tag{37}$$

The analogous estimation for $w$ instead of $v_2$ cannot hold in general, so that we consider

$$\left|\Pi(w)\sum_{\rho_k\in\mathfrak{B}_{13}}\frac{1}{w-\rho_k}\right| = \left|\sum_{\rho_k\in\mathfrak{B}_{13}}\frac{\Pi(w)}{w-\rho_k}\right|.$$

The last sum cannot exceed

$$32^{Q_1-1}\cdot(R\mu)^{Q_1-1}\cdot Q_1 \tag{38}$$

whereas

$$|\Pi(w)| < 32^{Q_1}(R\mu)^{Q_1}. \tag{39}$$

We obtain further

$$G(w) = \Pi(w)(\varphi(w)-Z) = \Pi(w)[\varphi(w)-\varphi(v_2)+\varphi(v_2)-Z] =$$
$$= \Pi(w)\left[\varphi_2(w)-\varphi_2(v_2)+\varphi_3(w)-\varphi_3(v_2)-\sum_{\rho_k\in\mathfrak{B}_{13}}\frac{1}{v_2-\rho_k}+\right.$$
$$\left.+\varphi(v_2)-Z+\theta\cdot 10C_8\ln D\right]+\Pi(w)[B_\Psi(w)-B_\Psi(v_2)]+\Pi(w)\sum_{\rho_k\in\mathfrak{B}_{13}}\frac{1}{w-\rho_k}. \tag{40}$$

Now, by (21): for $s=w$ and $s=v_2$

$$|B_\Psi(s)| < C_{12}\frac{\ln D}{\Psi(D)}\ln\Psi(D)\cdot\exp\left(0{,}1\frac{R}{\Psi(D)}\ln\Psi(D)\right). \tag{21'}$$

And, by definition of $v_2$ and $R$,

$$|\varphi(v_2)| = \left|\varphi\left(v_1+\frac{R}{4}\mu\right)\right| \leqslant M_{j_0+1} \leqslant \left(1-\frac{1}{(j_0+2)^2}\right)\cdot|Z|.$$

Taking into account (33), (36), (37), (38), (39), (40) and (21′), we get:

$$|G(w)|<$$
$$< 32^{Q_1}(R_\mu)^{Q_1}\left[10^9\cdot K_2\cdot|Z|+C_{12}\frac{\ln D}{\Psi(D)}\ln\Psi(D)\cdot\exp\left(0{,}1\,\frac{R}{\Psi(D)}\ln\Psi(D)\right)\right]+$$
$$+32^{Q_1}(R\mu)^{Q_1-1}. \qquad (41)$$

Whereas in the point $v_2$ we have by (40)

$$|G(v_2)|=|\Pi(v_2)|\cdot|\varphi(v_2)-Z|>|\Pi(v_2)|\cdot\frac{|Z|}{(j_0+2)^2}>(R\mu)^{Q_1}\cdot\frac{|Z|}{(j_0+2)^2}.$$

Hence

$$\left|\frac{G(w)}{G(v_2)}\right|\leqslant 32^{Q_1}[10^9(j_0+2)^2K_2+$$
$$+C_{12}\frac{(j_0+2)^2}{\Psi(D)}\ln\Psi(D)\ \exp\left(0{,}1\,\frac{R}{\Psi(D)}\ln\Psi(D)\right)\Big]+32^{Q_1}(j_0+2)^2 \qquad (42)$$

Hence

$$\left|\frac{G(w)}{G(v_2)}\right|\leqslant 32^{Q_1}\cdot 10^{10}(j_0+2)^2K_2\exp\left(0{,}1\,\frac{R}{\Psi(D)}\ln\Psi(D)\right). \qquad (43)$$

If $P$ is the number of zeros of $G(w)$ in the circle $\mathfrak{G}_1$, we shall have by Jensen's theorem

$$2^P\leqslant\sup\left|\frac{G(w)}{G(v_2)}\right|\quad\text{for } w\in\mathfrak{C}_2,$$

and so, by (43),

$$P\leqslant 2Q_1\ln 32+20\ln 10+4\ln(j_0+2)+2\ln K_2+\frac{R}{\Psi(D)}\ln\Psi(D). \qquad (44)$$

Now, $\ln(j_0+2)<\ln\ln R<R$, and, by Lemmas VIII and IV, $Q_1<8K_2\cdot 32R$, whence

$$P\leqslant 10^6K_2R. \qquad (45)$$

As the circle $\mathfrak{G}_1$ includes the circle $|s-v_1|\leqslant R\mu$ and $\varphi_1(s)$ has in this circle no more zeros than $G(s)$ we have proved Lemma XIV.

## § 17

Lemma XV. *The number of zeros and poles of $\varphi_1(s)$ in any circle of the type $|s-(1+iT)|\leqslant 0{,}4$ does not exceed $C_{15}\ln D(|T|+2)$.*

Proof. We have $\varphi_1(s)=\varphi(s)-Z=f(s)-B_\Psi(s)-Z=\frac{L'}{L}(s)-B_\Psi(s)-Z$; its number of zeros in that circle cannot surpass the number of zeros of the function

$$L(s)\varphi_1(s)=L'(s)-B_\Psi(s)L(s)-ZL(s).$$

For $s=2+iT$ we get easily $|L(s)\varphi_1(s)|>\frac{|Z|}{10}$. For $|w-(2+iT)|=\frac{13}{9}$ we get by (21) and by well known estimates for $L(w)$ and $L'(w)$, $|L(w)\varphi_1(w)|<D(|t|+2)$. Hence the lemma follows easily by Jensen's theorem, and Lemma IV.

§ 18

We have obtained sufficient information about the «density of zeros» of $\varphi(w)-Z$. We need now a lemma analogous to Lemma II, where some elementary properties of «almost periodicity» play a rôle, but it requires another fundamental estimate.

Lemma XVI. *Let $\Delta$ be a large number, $r_1<\Delta^{0,001}$ a large number, $\mu=\frac{1}{\Delta}$, $S(x)=\sum_{k=0}^{M-1}\frac{\exp(\rho_k-\rho_0)x}{(\rho_k-\rho_0+iT_1)^2}$, where $\rho_k$ are some points in the semi-circle $|s-\rho_0|\leqslant r_1\mu$, $\Re(s-\rho_0)\leqslant 0$, $1\leqslant M\leqslant(\ln\ln r_1)^2$; $\frac{1}{2}r_1^{0,1}\cdot\mu\leqslant T_1\leqslant r_1^{0,1}\cdot\mu$ and there are no $\rho_k$'s in $|t-T_1|\leqslant\frac{T_1}{M^2}$.*

*Then there exists a $\xi\in[10\Delta, 20\Delta]$ such that*

$$|S(\xi)|>\frac{1}{T_1^2}\exp(-\sqrt{\ln r_1}). \tag{46}$$

Proof. We consider $\Phi(x)=T_1^2S(x)=\sum_{k=0}^{M-1}\frac{T_1^2\exp(\rho_k-\rho_0)x}{(\rho_k-\rho_0+iT_1)^2}$. Putting

$\rho_k-\rho_0=z_k$; $\frac{T_1^2}{(\rho_k-\rho_0+iT)^2}=\Gamma_k$, we get the sum:

$$\Phi(x)=\sum_{k=0}^{M-1}\Gamma_k\exp z_k\cdot x,\quad \Re z_k\leqslant 0,\quad z_0=0.$$

We put now $\exp\left(z_k\cdot\frac{\Delta}{M}\right)=Z_k$; $Z_0=1$.

We shall prove now an auxiliary fact:

*There exists a $Z_{a_n}$ and a number $m\leqslant 100\ln M$, $m>0$, such that*

(1) *The ring $\exp(-M^{m+5})\leqslant|w-Z_{a_n}|\leqslant\exp(-M^m)$ is either entirely hollow, i. e. free from $Z_k$'s, or it contains only $Z_k$'s with $|\Gamma_k|\leqslant\leqslant\exp(-M^{m+5})$.*

(2) $|\Gamma_{a_n}|>\exp(-M^{m-5})$.

(3) $|Z_{a_n}-Z_0|=|Z_{a_n}-1|\leqslant\exp(-M^{20})$.

To this end, we begin the following process: We draw the circle $|w-Z_0|\leqslant\exp(-M^{20})$. If it contains no $Z_k$'s besides $Z_0$, we are already at our aim with $m=20$, and the process is over. If it is not so, we draw the circle $|w-1|\leqslant\exp(-M^{20\cdot 2})$. If the ring between these two circles is hollow, we are ready; if there is more points in the ring than in the inner circle, we carry on the process, till we shall arrive either at a circle $|w-1|\leqslant\exp(-M^{20k})$, such that there are no $Z_k$'s in the ring between it and the outer circle, $|w-1|\leqslant\exp(-M^{20(k-1)})$, or at a circle $|w-1|\leqslant\leqslant\exp(-M^{20k})$, containing no less points than the ring $\exp(-M^{20k})\leqslant\leqslant|w-1|\leqslant\exp(-M^{20(k-1)})$.

Here obviously $k \leqslant 2 \ln M$. In the first case we have achieved our aim; in the second case we consider the ring $\exp(-M^{20k-8}) \leqslant |w-1| \leqslant \exp(-M^{20k-15})$. If it contains only $Z_k$'s with $|\Gamma_k| \leqslant \exp(-M^{20k-10})$ we are at the end with $m=20k-15$; if it is not so, we take a point $Z_{\alpha_1}$ belonging to this ring, with $|\Gamma_{\alpha_1}| > \exp(-M^{20k-10})$ and draw the circle $|w-Z_{\alpha_1}| \leqslant \exp(-M^{20k})$. It contains no more than a half of points of the circle $|w-1| \leqslant \exp(-M^{20(k-1)})$.

If it is hollow, or contains only points $Z_k$ with $|\Gamma_k| < \exp(-M^{20k+10})$ we are ready. If it is not so, we carry on our process with $Z_{\alpha_1}$ instead of $Z_0 = 1$. Thus we shall arrive at a ring $\exp(-M^{20l-8}) \leqslant |w-Z_{\alpha_1}| \leqslant \exp(-M^{20l-15})$ containing no more points than half of their number in the including circle; $l > k$. If it is hollow, or contains only points $Z_k$ with $|\Gamma_k| < \exp(-M^{20(l-1)+10})$, we are ready with $m = 20l - 16 < 2 \ln M$, since

$$20k - 10 \leqslant 20(l-1) - 10 = 20l - 30 \leqslant m - 5,$$

$$20(l-1) + 10 = 20l - 10 = m + 5.$$

If it is not so, we carry on our process till we have found the required $Z_{\alpha_n}$. The quantity of steps is obviously $< 2 \ln M$ since each step at least halves the number of points in the corresponding rings.

We put now

$$Y_k = \exp(z_k \cdot 10\Delta), \quad Y_0 = 1;$$

$$x_0 = 10\Delta, \ x_1 = x_0 + \frac{\Delta}{M}, \ldots, \ x_{M-1} = x_0 + \frac{M-1}{M}\Delta.$$

We obtain

$$\left.\begin{aligned}
\Phi(x_0) &= \sum_{k=0}^{M-1} \Gamma_k Y_k = \nu_0, \\
\Phi(x_1) &= \sum_{k=0}^{M-1} \Gamma_k Y_k Z_k = \nu_1, \\
\Phi(x_2) &= \sum_{k=0}^{M-1} \Gamma_k Y_k Z_k^2 = \nu_2, \\
& \cdots\cdots\cdots\cdots\cdots \\
\Phi(x_{M-1}) &= \sum_{k=0}^{M-1} \Gamma_k Y_k Z_k^{M-1} = \nu_{M-1}.
\end{aligned}\right\} \qquad (47)$$

Let us put now $Z_{\alpha_n} = Z_l$. How large is the error, if we identify all the points $Z_k$ in the circle $|w - Z_l| \leqslant \exp(-M^{m+5})$?—We have

$$|Z_k| \leqslant 1, \quad |Z_k^q - Z_l^q| = |Z_k - Z_l| \cdot |Z_k^{q-1} + Z_k^{q-2} Z_l + \ldots + Z_l^{q-1}|.$$

As $|Z_k - Z_l| \leqslant \exp(-M^{m+5})$, we have
$|Z_k^q - Z_l^q| \leqslant 10M \cdot \exp(-M^{m+5}) < \exp\left(-\frac{1}{2}M^{m+5}\right) \quad (q = 1, 2, \ldots, 10M)$.
Now, by definition of $T_1$:

$$|\Gamma_j| = \left|\frac{T_1^2}{(\rho_k - \rho_0 + iT_1)^2}\right| < \frac{T_1^2}{T_1^2/10M^2} \leqslant 10M^2.$$

Hence the process of identification and deleting terms with $|\Gamma_j| \leqslant \leqslant \exp(-M^{m+5})$ will cause an error not exceeding

$$10M^2 \cdot M \cdot \exp\left(-\frac{1}{2}M^{m+5}\right) < \exp\left(-\frac{1}{4}M^{m+5}\right).$$

Putting now

$$Z_1' = Z_n, \quad \Gamma_1' = \Gamma_n, \quad \nu_k' = \nu_k + \theta \cdot \exp\left(-\frac{1}{4}M^{m+5}\right)$$

we obtain, after this process, the system of equations:

$$\left.\begin{aligned} &\sum_{j=1}^{M_1} \Gamma_j' Y_j' m_j \cdot 1 = \nu_0', \\ &\sum_{j=1}^{M_1} \Gamma_j' Y_j' m_j \cdot Z_j = \nu_1', \\ &\sum_{j=1}^{M_1} \Gamma_j' Y_j' m_j \cdot Z_j^{M_1-1} = \nu_{M_1-1}'. \end{aligned}\right\} \tag{48}$$

Here $M_1 < M$ so that superflous equations are cancelled. $m_j$ equals to the number of identified points for $j=1$ and $\geqslant 1$ for $j > 1$

$$|Y_1'| = |Z_1'^{10M}| > (1 - \exp(-M^{20}))^{10M} > \frac{1}{2}.$$

We obtain now, resolving the system of equations (48) with respect to $\Gamma_1' Y_1' m_1$:

$$\Gamma_1' Y_1' m_1 = \frac{A_1}{A}, \tag{49}$$

$$A_1 = \begin{vmatrix} \nu_0' & 1 & \ldots & 1 \\ \nu_1' & Z_2' & \ldots & Z_{M_1}' \\ \cdot & \cdot & \cdot & \cdot \\ \nu_{M_1-1}' & Z_2'^{M_1-1} & \ldots & Z_{M_1}'^{M_1-1} \end{vmatrix}, \qquad A = \begin{vmatrix} 1 & \ldots & 1 \\ Z_1' & \ldots & Z_{M_1}' \\ \cdot & \cdot & \cdot \\ Z_1'^{M_1-1} & \ldots & Z_{M_1}'^{M_1-1} \end{vmatrix},$$

$$\left|\frac{A_1}{A}\right| \leqslant \frac{\sum_{l=0}^{M_1-1} |D_{M_1-1-l}| \, |\nu_l'|}{|Z_1' - Z_2'| \cdot |Z_1' - Z_3' \ldots |Z_1' - Z_{M_1}'|},$$

where $D_{M_1-1-l}$ is the sum of all possible products of $M_1 - 1 - l$ quantities chosen between $Z_2', \ldots, Z_{M_1}'$.

Let $\max|\nu_l'| = \overline{|\nu|}$, then we have, as $|Z_j'| \leqslant 1$:

$$\sum_{l=0}^{M_2-1} |D_{M_1-1-l}| \, |\nu_l'| < 4^M \overline{|\nu|}, \tag{50}$$

$$|Z_1' - Z_2'| \ldots |Z_1' - Z_{M_1}'| > \exp(-M^m M) = \exp(-M^{m+1}).$$

Whence $|\Gamma_1' Y_1' m_1| < 4^M \overline{|\nu|} \exp(M^{m+1})$, as $|Y_1'| > \frac{1}{2}$, $|\Gamma_1'| \geqslant \exp(-M^{m-5})$, we have

$$m_1 < \overline{|\nu|} \cdot \exp(4M^{m+1}). \tag{51}$$

Suppose now that in (47) all $|\nu_j|$'s do not exceed $\exp(-M^{m+5})$. Then, as $\nu_j' = \nu_j + \theta \cdot \exp\left(-\frac{1}{4} M^{m+5}\right)$, we have $\overline{|\nu|} \leqslant 2\exp\left(-\frac{1}{4} M^{m+5}\right)$, and so $m_1 < 2\exp\left(-\frac{1}{8} M^{m+5}\right)$, which is impossible, since $m_1$ is an integer $\geqslant 1$.

Hence we have for some $x_k$ in (47):

$$|\Phi(x_k)| \geqslant \exp(-M^{m+5}).$$

Now,

$$m+5 \leqslant 200 \ln M; \quad M^{m+5} \leqslant \exp(200 \ln^2 M) < \exp(800 \ln\ln\ln r_1)^2) <$$

$$< \exp\left(\frac{1}{2} \ln\ln r_1\right) = \sqrt{\ln r_1}$$

or $r_1$ sufficiently large, and so

$$|\Phi(x_k)| \geqslant \exp(-\sqrt{\ln r_1}), \quad |S(x_k)| \geqslant \frac{1}{T_1^2} \exp(-\sqrt{\ln r_1}).$$

This is (46), and the lemma is proved, with $\xi = x_k$.

## § 19

Lemma XVII. $\varphi_1(s) = \varphi(s) - Z$ *has no zeros in* $\sigma \geqslant 1 + \mu$.

Proof. We have there $|f(s)| < C_8 \ln D$, $B_\Psi(s) < C_8 \ln D$ and as $|Z| > 4C_8 \ln D$, the proof is immediate.

Lemma XVIII. *In the stripe* $0{,}6 \leqslant \sigma \leqslant 0{,}65$ *there exists an infinite locally rectifiable contour* $\mathfrak{C}$ *under condition:*

(1) $Q(w) = \frac{\varphi_1'(w)}{\varphi_1(w)}$ *is of order* $\ln^4 D(|t|+2)$ *on* $\mathfrak{C}$.

(2) *The length of the portion of* $\mathfrak{C}$ *between* $T$ *and* $T+1$ *does not exceed* $\ln^2 D(|t|+2)$.

Proof. This is an easy consequence of Lemmas I and XV. Lemma I should be adapted to meromorphic $\varphi_1(s)$.

## § 20

On the line $\sigma=2$ we have $|Q(w)|=\left|\frac{\varphi'(w)}{\varphi(w)-Z}\right|<\frac{1}{\ln D}$. Let $\tau_1$ be any real number of the segment $[Iv-2, Iv+2]$ with $|\tau_0-\tau_1|\geqslant 0{,}1$ and $\delta\in(0,1)$; we introduce the integral

$$\Phi(x,\chi,Z,\tau_0,\tau_1)=\Phi(x,\chi)=\frac{1}{2\pi i}\int_{2-i\infty}^{2+i\infty}\delta^{-w}\Gamma(w-i\tau_1)Q(w)\,dw \tag{52}$$

which is absolutely convergent. Here $x=-\ln\delta$, as in § 9.

Moving the contour to the line $\mathfrak{C}$ of Lemma XVIII (which is legitime), we pass through the poles $\rho_k$ of $Q(w)$ and thus we obtain

$$\Phi(x,\chi,Z,\tau_0,\tau_1)=\sum_k \pm m_k\Gamma(\rho_k-i\tau_1)\delta^{-\rho_k}+R_{\mathfrak{C}}. \tag{53}$$

Here $m_k$ denotes the multiplicity of the corresponding poles, i. e. that of the corresponding zero of $\varphi(w)-Z$ or $L(w,\chi)$; the signs $(+)$ refer to the zeros of $\varphi(w)-Z$, and the signs $(-)$ to those of $L(w,\chi)$.

$$|R_{\mathfrak{C}}|=\left|\frac{1}{2\pi i}\int_{\mathfrak{C}}\delta^{-w}\Gamma(w-i\tau_1)Q(w)\,dw\right|<\delta^{-0,65}\ln^{10}D, \tag{54}$$

by Lemma XVIII.

The point $v$ of § 16, with $\varphi(v)=Z$, is evidently one of our $\rho_k$'s; we denote it $\rho_0=\alpha_0+i\gamma_0$.

By what precedes, there are no $\rho_k$'s in $\sigma>\alpha_0$; $|t-\gamma_0|\leqslant(\Psi(D))^{45}$ and in $\sigma>1+\mu$, moreover, all $\rho_k$'s situated in $\sigma>\alpha_0-2r_1\mu$; $|t-\tau_0|\leqslant\leqslant(\Psi(D))^{45}$ have the sign $(+)$ in (53). The distribution of other $\rho_k$'s is explained in Lemma XV.

Denoting $\rho_k-\rho_0=\rho_k-(\alpha_0+i\gamma_0)=-\sigma_k+it_k$, so that $-\sigma_k\leqslant 0$ for $|t_k|\leqslant\leqslant(\Psi(D))^{45}$, $-\sigma_0+it_0=0$, $-\sigma_k\leqslant\frac{2\Psi(D)}{\ln D}$ always; $\Gamma_j=\Gamma(\rho_j-i\tau_1)$ so that $|\Gamma_j|\leqslant 10^3\exp(-|t_j|)$, we obtain, since $x=-\ln\delta$,

$$\begin{aligned}\Phi(x,\chi,Z,\tau_0,\tau_1)&=e^{\alpha_0x}\cdot\sum_{k=-\infty}^{\infty}\pm m_j\Gamma_j\exp(-\sigma_j+it_j)x+R_{\mathfrak{C}}=\\&=S(x,\chi,Z,\tau_0,\tau_1)e^{\alpha_0x}+R_{\mathfrak{C}}.\end{aligned} \tag{55}$$

## § 21

We come now to the crucial Lemma XIX, analogous to the estimate (11) of § 9 and the conditional estimate (5) of the paper [1].

Lemma XIX. *For $\Psi(D)>K_{11}$, there are two constants $K_{12}$ and $K_{13}$ depending upon $K_1,\ldots,K_{10}$ only such that $K_{13}>K_{12}>10$ and that for $X_1=K_{12}\ln D$, $X_2=K_{13}\ln D$ we have*

$$\int_{X_1}^{X_2}|S(x,\chi,Z,\tau_0,\tau_1)|^2\,dx>\frac{\ln D}{(\Psi(D))^2}. \tag{56}$$

Proof. We shall distinguish two cases:

Case I. The number $2^{j_0}\Delta_1$ of the Lemma XIII satisfies a system of $\leqslant C_{17}$ inequalities of the type $2^{j_0}\Delta_1 > A_n(K_0, K_1, \ldots, K_{10}, C_\alpha)$, which will be introduced in what follows

Case II.

$$2^{j_0}\Delta_1 < A_n(K_0, K_1, \ldots, K_{10}, C_\alpha). \tag{57}$$

Let $10^{-5}\Delta_1 = r_1$, $r_1$ being the number of Lemma XVI, and let $(\ln\ln r_1)^2 > K_8 \ln(j_0+2) = K_8 C_{18} \ln\ln r_1$ (cf. Lemma XIII).

Taking $\Delta = \ln D$ in the Lemma XVI, we draw the circle $|s-\rho_0| \leqslant r_1\mu$ satisfying the conditions of that Lemma; we denote it $\mathfrak{G}$ and we fix the number $T_1$ (cf. Lemma XVI). We put now

$$S(x, \chi, Z, \tau_0, \tau_1) \cdot \exp(iT_1 x) = \Psi_1(x) = \\ = \sum_j \pm m_j \Gamma_j \exp(-\sigma_j + i(t_j + T_1))\, x. \tag{58}$$

Taking now $x_1 = 10 \ln D$ we obtain, integrating over $[x_1, x]$:

$$\int_{x_1}^{x} \Psi_1(x)\,dx = \sum_j \pm m_j \Gamma_j \frac{\exp(-\sigma_j + i(t_j + T_1))\,x}{-\sigma_j + i(t_j + T_1)} + A_0 = \Psi_2(x) + A_0.$$

Repeating the integration we obtain:

$$\int_{x_2}^{x} (\Psi_2(x) + A_0)\,dx = \int_{x_1}^{x} (x-y)\,\Psi_1(y)\,dy =$$

$$= \sum_j \pm m_j \Gamma_j \frac{\exp(-\sigma_j + i(t_j + T_1))x}{(-\sigma_j + i(t_j + T_1))^2} + A_0(x - x_1) = \Psi_2(x) + A_0(x - x_1). \tag{59}$$

We shall prove now an auxiliary

Lemma XX. *If* $\mathfrak{G}$ *denotes the circle* $|s-\rho_0| \leqslant r_1\mu$ *and* $x \in [10 \ln D, (\Psi(D))^{35} \ln D]$, *then*

$$\left| \sum_{\rho_j \in \mathfrak{G}} \pm m_j \Gamma_j \frac{\exp(-\sigma_j + i(t_j + T_1))\,x}{(-\sigma_j + i(t_j + T_1))^2} \right| < \frac{C_{19} K_{10} \ln^2 D}{r_1}. \tag{60}$$

Proof. For $-\sigma_j < -\dfrac{\ln\ln D}{\ln D}$ the sum of corresponding terms is, by Lemma XV, $\ll (\ln D)^{-10} \cdot \dfrac{1}{\mu^2} < (\ln D)^{-7}$. The sum of terms with $|t_j| > (\Psi(D))^{40}$ cannot exceed $C_{20} \ln D \cdot \exp\left(-\frac{1}{2}(\Psi(D))^{40}\right)$. In fact, as $-\sigma_j \leqslant \dfrac{2\Psi(D)}{\ln D}$ and $|\Gamma_j| < 1\,000 \exp(-|t_j|)$ each separate term with the index $j$ cannot exceed

$$1000 \exp(-|t_j|) \cdot \exp(2(\Psi(D))^{36}), \qquad |t_j| \geqslant (\Psi(D))^{40}$$

and hence the estimation follows easily by Lemma XV. For the remaining terms, $-\sigma_j \leqslant 0$; the quantities of corresponding $\rho_j$'s in the circles $|w-\rho_0| \leqslant R\mu$ with $R \in [4r_1, \ln\ln D]$ is $< K_{10} R$ by Lemma XIV. Hence drawing the circle $|s-\rho_0| \leqslant 2^k r_1 \cdot \mu$, we obtain the following estimate for their sum:

$$10 \cdot \sum_{k=1}^{\infty} \frac{2^k \cdot K_{10} r_1}{2^{2k} \cdot r_1^2 \cdot \mu^2} < 20 \cdot \frac{K_{10} \ln^2 D}{r_1}.$$

Summing up these results, we obtain (60).

We consider now the sum of terms with $|\rho_k-\rho_0| \leqslant r_1\mu$, i. e. $\rho_k \in \mathfrak{G}$. We shall now prove that

$$\sum_{\rho j \in \mathfrak{G}} m_j \Gamma_j \frac{\exp(-\sigma_j + i(t_j+T_1))\xi}{(-\sigma_j + t(t_j+T_1))^2} > \frac{1}{2T_1^2} \exp(-\sqrt{\ln r_1}) \tag{61}$$

for a certain $\xi \in [10 \ln D, 20 \ln D]$. In fact, in this circle we have $|\Gamma_j - \Gamma_0| < C_{21} r_1 \mu$.

Denoting $\sum m_j = M$ we have, by Lemma XVI,

$$\left| \sum_{\rho k \in \mathfrak{G}} m_j \frac{\exp(-\sigma_j + i(t_j+T_1))\xi}{(-\sigma_j + i(t_j+T_1))^2} \right| > \frac{\exp(-\sqrt{\ln r_1})}{T_1^2}.$$

The difference of this sum and that of (61) cannot exceed $C_{21} r_1 \mu \cdot \frac{M}{T_1^2} \cdot M = \frac{1}{T_1^2} \cdot \frac{C_{21} r_1 M^3}{\ln D} < \frac{1}{T_1^2 \sqrt{\ln D}}$ as $r_1 < (\ln D)^{0{,}001}$ and hence we obtain (61).

Now, as $|T_1| \leqslant r_1^{0{,}1}\mu$, we obtain, juxtaposing (60) and (61), for $r_1$ surfficiently large with respect to $K_{10}$:

$$|\Psi_2(\xi)| = \left| \sum_j \Gamma_j \frac{\exp(-\sigma_j + i(t_j+T_1))\xi}{(-\sigma_j + i(t_j+T_1))^2} \right| > \frac{\exp(-\sqrt{\ln r_1})}{3T_1^2}. \tag{62}$$

Here $T_1 \in \left[\frac{1}{2} r_1^{0{,}1}\mu,\ r_1^{0{,}1}\mu\right]$.

Returning now to (59), we distinguish two cases:

(1) $|\Psi_2(\xi) + A_0(\xi - x_1)| \leqslant \frac{1}{6T_1^2} \exp(-\sqrt{\ln r_1})$;

(2) $|\Psi_2(\xi) + A_0(\xi - x_1)| > \frac{1}{6T_1^2} \exp(-\sqrt{\ln r_1})$.

In case 1 we obtain, making use of (62):

$$|A_0| \geqslant \frac{1}{6T_1^2} \frac{\exp(-\sqrt{\ln r_1})}{(\xi - x_1)} > \frac{1}{10^3} \frac{\ln D \cdot \exp(-\sqrt{\ln r_1})}{r_1^{0{,}2}}. \tag{63}$$

We obtain now: for $x \in [x_1, (\Psi(D))^{35} \cdot \ln D]$

$$\left| \int_{x_1}^{x} A_0(x-x_1)\,dx \right| = \left| A_0 \cdot \frac{(x-x_1)^2}{2} \right| >$$

$$> 2 \cdot 10^{-3} \cdot |x-x_1|^2 \cdot \frac{\ln D \cdot \exp(-\sqrt{\ln r_1})}{r_1^{0{,}2}}, \tag{64}$$

$$\left| \int_{x_1}^{x} \Psi_2(x)\,dx \right| \leqslant \frac{C_{22} \cdot K_{10} \ln^3 D}{r_1^3} + \frac{M^4}{T_1^3} < \frac{\ln^3 D}{r_1^{0{,}29}} \tag{65}$$

(for $r_1$ large with respect to $K_{10}$).

Combining (64) with (65), we obtain for $X_3 = 20 \ln D$

$$\int_{x_1}^{X_3} |\Psi_2(x) + A_0(x - x_1)| \, dx \geqslant \left| \int_{x_1}^{X_3} \Psi_2(x) \, dx + \int_{x_1}^{X_3} A_0(x - x_1) \, dx \right| \geqslant$$

$$\geqslant 10^{-4} \cdot \frac{\ln^3 D \cdot \exp(-\sqrt{\ln r_1})}{r_1^{0,2}}.$$

Hence, there exists an $X_4 \in [x_1, X_3] \subset [10 \ln D, 20 \ln D]$ such that

$$|\Psi_2(X_4) + A_0(X_4 - x_1)| \geqslant 10^{-6} \cdot \frac{\ln^2 D \cdot \exp(-\sqrt{\ln r_1})}{r_1^{0,2}}. \tag{66}$$

Hence, from (59):

$$\left| \int_{x_1}^{X_4} (X_4 - y) \Psi_1(y) \, dy \right| \geqslant 10^{-6} \cdot \frac{\ln^2 D \cdot \exp(-\sqrt{\ln r_1})}{r_1^{0,2}}. \tag{67}$$

Now, $r_1 \leqslant 2\Psi(D)$. Hence $\frac{\exp(-\sqrt{\ln r_1})}{r_1^{0,2}} < \frac{1}{(\Psi(D))^{0,5}}$, $\Psi(D)$ being large enough, and so

$$\left| \int_{x_1}^{X_4} (X_4 - y) \Psi_1(y) \, dy \right| \geqslant 10^{-6} \cdot \frac{\ln^2 D}{(\Psi(D))^{0,5}}. \tag{68}$$

Hence, by Schwarz's inequality,

$$\int_{x_1}^{X_4} |X_4 - y|^2 \, dy \cdot \int_{x_1}^{X_4} |\Psi_1(y)|^2 \, dy \geqslant \left| \int_{x_1}^{X_4} (X_4 - y) \Psi_1(y) \, dy \right|^2 \geqslant 10^{-12} \cdot \frac{\ln^4 D}{\Psi(D)}.$$

Now

$$\int_{x_1}^{X_4} |X_4 - y|^2 \, dy < 10^4 \ln^3 D,$$

so that

$$\int_{x_1}^{X_4} |\Psi_1(y)|^2 \, dy \geqslant 10^{-16} \frac{\ln D}{\Psi(D)} > \frac{\ln D}{(\Psi(D))^2} \tag{69}$$

for $\Psi(D) > K_{11}(K_{10})$. As $|\Psi_1(y)| = |S(y, \chi, Z, \tau_0, \tau_1)|$ this is (56).

In case 2 we obtain at once (66) and draw the same conclusions, leading to (69).

We have thus completed the analysis of the case I and have to consider the case II (relation (57)), when $r_1 = 10^{-5} 2^{j_0} \Delta_1 < A_1(K_{10})$.

We return to the original function

$$S(x, \chi, Z, \tau_0, \tau_1) = \sum_{j=-\infty}^{\infty} \pm m_j \Gamma_j \exp(-\sigma_j + it_j) x.$$

By (57), Lemmas XIII and XIV, any circle with the center on $\mathfrak{S}[\alpha_0;\ |t-\gamma_0|\leq(\Psi(D))^{45}]$ and the radius $r\mu$ with $r\in[1,\ \ln\ln D]$ contains no more than $A_2(K_{10}, K_9, \ldots, K_1, K_0)\cdot r$ $\rho_j$'s while for $|t_j|\geq(\Psi(D))^{45}$ we have $-\sigma_j'\leq 2\Psi(D)\cdot\mu$ and the sign $(+m_j)$ for $\sigma_j<K_1\mu$; the distribution of $\rho_j$'s with $|t_j|\geq(\Psi(D))^{45}$ being described by Lemma XVII.

Under these circumstances, the estimate (56) follows easily by the method of iterated integration, which is given in detail in the paper [3] (§§ 11, 12, 13, 14, 15); (56) is even weaker than the estimate (13), proved there for an analogous function.

## § 22

We have by (52)

$$\Phi(x,\ \chi,\ Z,\ \tau_0,\ \tau_1)=\frac{1}{2\pi i}\int\limits_{2-i\infty}^{2+i\infty}\delta^{-w}\Gamma(w-i\tau_1)Q(w)\,dw$$

with $Q(w)=\frac{\varphi'(w)}{\varphi(w)-Z}$.

On the line $\sigma=2$ we have $\frac{|\varphi(w)|}{Z}<\frac{1}{\ln D}$ and hence, for $m>\ln D$, $\left|\frac{\varphi(w)}{Z}\right|^m<\exp(-\ln D\cdot\ln\ln D)$.

Now on $\sigma=2$

$$Q(w)=-\frac{\varphi'(w)}{Z-\varphi(w)}=-\frac{\varphi'(w)}{Z}\left(1+\frac{\varphi(w)}{Z}+\frac{(\varphi(w))^2}{Z^2}+\ \ldots\right)=$$
$$=Q_1(w)+\theta\cdot\exp(-\ln D\cdot\ln\ln D) \tag{70}$$

with

$$Q_1(w)=\frac{\varphi'(w)}{Z}\sum_{m=0}^{\ln D}\frac{(\varphi(w))^m}{Z^m}. \tag{71}$$

Substituting this expression in (52), we obtain for $\delta^{-1}<\exp(\ln D(\ln\ln D)^{1/2})$

$$\left.\begin{aligned}&\Phi(x,\chi,Z,\tau_0,\tau_1)=\frac{1}{2\pi i}\int\limits_{2-i\infty}^{2+i\infty}\delta^{-w}\cdot\Gamma(w-i\tau_1)Q_1(w)\,dw+R_1,\\&R_1=\theta\exp\left(-\frac{1}{2}\ln D\cdot\ln\ln D\right).\end{aligned}\right\} \tag{72}$$

## § 23

We wish now to expand $Q_1(w)$ in Dirichlet's series converging absolutely for $\sigma>1$. To this end, we note that in this half-plane

$$\varphi(w)=-\frac{1}{w-i\tau_0}\sum_{p\geq N_\Psi}\frac{\chi(p)\ln p}{p^w}$$

whence, for $m\geq 0$, integer

$$(\varphi(w))^{m+1}=\frac{(-1)^{m+1}}{(w-i\tau_0)^{m+1}}\sum_{n=2}^{\infty}\frac{\chi(n)V_{m+1}(n)}{n^w}, \tag{73}$$

where

$$V_{m+1}(n)=\sum_{p_1\cdots p_{m+1}=n;\ p_j\geqslant N_\Psi}\ln p_1\ln p_2\cdots\ln p_{m+1}$$

and the sum is extended over all possible permutations of the $p_j$'s, each of them being $\geqslant N_\Psi$. The number of these permutations is $\Gamma(m+2)$. By the well known theorem on the arithmetic and geometric means, we have

$$\ln p_1\cdots\ln p_{m+1}\leqslant\left(\frac{\ln p_1+\cdots+\ln p_{m+1}}{m+1}\right)^{m+1}=\frac{(\ln n)^{m+1}}{(m+1)^{m+1}}.$$

Hence, by Stirling's theorem, we obtain

$$V_{m+1}(n)<\frac{(\ln n)^{m+1}}{(m+1)^{m+1}}\Gamma(m+2)<10^4\cdot(\ln n)^{m+1}e^{-\frac{m}{2}}. \tag{74}$$

We have now in (71)

$$\frac{\varphi'(w)}{Z}\cdot\frac{(\varphi(w))^m}{Z^m}=\frac{1}{Z^{m+1}}\cdot\frac{1}{m+1}\cdot\frac{d}{dw}\{(\varphi(w))^{m+1}\},$$

$$\frac{d(\varphi(w))^{m+1}}{dw}=(-1)^m(m+1)\frac{1}{(w-i\tau_0)^{m+2}}\sum_{n=2}^{\infty}\frac{\chi(n)V_{m+1}(n)}{n^w}+$$

$$+(-1)^m\cdot\frac{1}{(w-i\tau_0)^{m+1}}\sum_{n=2}^{\infty}\frac{\chi(n)V_{m+1}(n)\ln n}{n^w},$$

so that

$$\frac{\varphi'(w)(\varphi(w))^m}{m+1}=(-1)^m\cdot\frac{1}{(w-i\tau_0)^{m+2}}\sum_{n=2}^{\infty}\frac{\chi(n)V_{m+1}(n)}{n^w}+$$

$$+(-1)^m\cdot\frac{1}{m+1}\cdot\frac{1}{(w-i\tau_0)^{m+1}}\sum_{n=2}^{\infty}\frac{\chi(n)V_{m+1}(n)\ln n}{n^w}. \tag{75}$$

## § 24

We are thus led to the study of Mellin's transform

$$E(y,k)=\frac{1}{2\pi i}\int_{2-i\infty}^{2+i\infty}y^{-w}\cdot\frac{\Gamma(w-i\tau_1)}{(w-i\tau_0)^k}\,dw$$

for $y>0$, which embodies the «method of summation» in (52).

L e m m a XXI. *The function* $E(y,k)$ *satisfies the following inequalities:*

*For* $y\leqslant 1$, *if* $|\ln y|>C_{24}$,

$$|E(y,k)|<|\ln y|^k\cdot\max k\cdot\left\{\frac{1}{\Gamma\left(\frac{k}{2}\right)},\ \left(\frac{C_{23}}{|\ln y|}\right)^{k/2}\right\} \tag{76}$$

$$|E(y,k)|<C_{25}^k\ \textit{if}\ |\ln y|\leqslant C_{24}.$$

*For* $1 \leqslant y \leqslant D$, $k \leqslant 2 \ln D$,

$$|E(y, k)| \leqslant e^{-\frac{y}{4}} e^{C_{26} k \ln k}. \tag{77}$$

*For* $y \geqslant D$, $k \leqslant 2 \ln D$,

$$|E(y, k)| < e^{-10^{-3} y}. \tag{78}$$

Proof. We consider firstly

$$E(y, k) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} y^{-w} \Gamma(w - i\tau_1) \frac{dw}{(w - i\tau_0)^k}$$

for $0 < y \leqslant 1$, $y^{-1} \geqslant 1$. The poles of the integrands are $w = i\tau_0$ (multiple pole), and $w = i\tau_1 - n$, $n = 0, 1, 2, \ldots$ (simple poles). Drawing the semi-circles $|w - i\tau_1| = \frac{2N+1}{2}$, $N$ integer $\to \infty$, and moving the contour to them, we obtain that $E(y, k)$ equals to the sum of residues in $\Re w \leqslant 0$. In the vicinity of $w = i\tau_1$, we have

$$\frac{\Gamma(w - i\tau_1)}{(w - i\tau_0)^k} = \frac{F(w)}{(w - i\tau_0)^k} = \frac{1}{(w - i\tau_0)^k} \Big( F(i\tau_0) + (w - i\tau_0) \frac{F'(i\tau_1)}{1!} + \\ + \ldots + \frac{(w - i\tau_1)^k \cdot F^{(k)}(i\tau_1)}{k!} + \ldots \Big),$$

$$y^{-w} = y^{-i\tau_0} \cdot y^{-(w - i\tau_0)} = y^{-i\tau_0} \Big( 1 + \frac{(w - i\tau_0)}{1!} (-\ln y) - \frac{(w - i\tau_0)^2}{2!} (-\ln y)^2 + \\ + \ldots + \frac{(w - i\tau_0)^k}{k!} (-\ln y)^k + \ldots \Big).$$

Hence the residue in $w = i\tau_1$ is

$$y^{-i\tau_0} \left\{ \frac{(-\ln y)^{k-1}}{(k-1)!} F(i\tau_0) + \frac{(-\ln y)^{k-2}}{(k-2)!} \frac{F'(i\tau_0)}{1!} + \ldots + \frac{F^{(k-1)}(i\tau_0)}{1!\,(k-1)!} \right\}.$$

As $|\tau_1 - \tau_0| > 0{,}1$, we conclude easily by Cauchy's theorem

$$\left| \frac{F^{(m)}(i\tau_0)}{m!} \right| < C_{27}^m.$$

The residues in $w = i\tau_1 - n$, $|\tau_1 - \tau_0| \geqslant 0{,}1$, give the sum

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\Gamma(n+1)} y^{n - i\tau_1} \cdot \frac{1}{(i(\tau_1 - \tau_0) - n)^k}, \qquad |\tau_1 - \tau_0| \geqslant 0{,}1.$$

Summing up these estimates, we easily obtain (76).

Let now $y > 1$; then we apply the theorem on the «Faltung» of Mellin transforms [7]; under certain conditions, which are satisfied in our case, we have: if

$$\Phi_1(x) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} x^{-w} \varphi_1(w)\, dw, \quad \Phi_2(x) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} x^{-w} \varphi_2(w)\, dw,$$

$$\Phi_{1,2}(x) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} x^{-w} \varphi_1(w) \varphi_2(w)\, dw,$$

then

$$\Phi_{1,2}(y)=\int_0^\infty \Phi_1(u)\Phi_2\left(\frac{y}{u}\right)\frac{du}{u}.$$

In the case in point, we have

$$E(y,k)=\frac{1}{2\pi i}\int_{2-i\infty}^{2+i\infty} y^{-w}\frac{\Gamma(w-i\tau_1)}{(w-i\tau_0)^k}\,dw=\frac{y^{-i\tau_0}}{2\pi i}\int_{2-i\infty}^{2+i\infty} y^{-w}\frac{\Gamma(w+i\tau_0-i\tau_1)}{w^k}\,dw.$$

Taking here

$$\varphi_1(w)=\Gamma(w+i\tau_0-i\tau_1),\quad \varphi_2(w)=\frac{1}{w^k},$$

we have for $0<u_1<1$, $\Phi_2(u_1)=0$ $(u_1\geqslant 0)$,

$$\Phi_1(u)=u^{i(\tau_0-\tau_1)}\cdot e^{-u},\quad \Phi_2(u_1)=\frac{(-\ln u_1)^{k-1}}{(k-1)!}.$$

Hence we obtain

$$|E(y,k)|\leqslant\frac{1}{(k-1)!}\int_y^\infty e^{-u}\left|\ln\frac{y}{u}\right|^{k-1}\frac{du}{u}.$$

For $1\leqslant y\leqslant D$, $k\leqslant\ln D$, we have

$$|E(y,k)|\leqslant e^{-\frac{y}{2}}\cdot\frac{1}{(k-1)!}\int_y^\infty e^{-\frac{u}{2}}\left|\ln\frac{y}{u}\right|^{k-1}du.$$

And hence (77) and (78) may be easily derived.

## § 25

Combining (73), (71) and (75), we obtain

$$\Phi(x,\chi,Z,\tau_0,\tau_1)=\sum_{m=0}^{\ln D}\frac{(-1)^m}{Z^{m+1}}\sum_{n=2}^{\infty}\chi(n)\cdot\left\{\frac{V_{m+1}(n)\ln n}{m+1}E(\delta n,m+1)+\right.$$
$$\left.+V_{m+1}(n)E(\delta n,m+2)\right\}+R_1. \tag{79}$$

Here $|R_1|<\exp\left(-\frac{1}{2}\ln D\ln\ln D\right)$ provided that $\delta^{-1}<\exp(\ln D\cdot$ $\cdot(\ln\ln D)^{1/2})$.

As $|Z|>\ln D$ we have obviously

$$|\Phi(x,\chi,Z,\tau_0,\tau_1)|\leqslant\sum_{m=0}^{\ln D}\frac{1}{(\ln D)^{m+1}}\left|\sum\nolimits_m\right|+|R_1|, \tag{80}$$

where

$$\sum\nolimits_m=\sum_{n=2}^{\infty}\chi(n)\left\{\frac{V_{m+1}(n)\ln n}{m+1}E(\delta n,m+1)+\right.$$
$$\left.+V_{m+1}(n)E(\delta n,m+2)\right\}. \tag{81}$$

If $\delta^{-1}$ is not too large, we may omit certain terms $\sum_m$ with a negligible error. In point of fact, let

$$\delta^{-1} < D^{K_{14}} = \exp(K_{14} \ln D) \tag{82}$$

where $K_{14} = K_{13}^{1000}$, $K_{13}$ being the constant of Lemma XIX, where $X_2 = K_{13} \ln D$.

Lemma XXII. *For* $m \geq M_\Psi = 10^4 K_{14} \frac{\Psi(D)}{\ln D}$, $m \leq \ln D$, *we have*

$$\left|\sum_m\right| < \exp(-10^{-5} D) \ \textit{for}\ \delta^{-1} < D^{K_{14}}. \tag{83}$$

Proof. We have by (19), $N_\Psi = \exp\left(0{,}001 \frac{\ln D}{\Psi(D)} \ln \Psi(D)\right)$. Hence, $N_\Psi^m > N_\Psi^{M_\Psi} > \exp(10 K_{14} \ln D)$. Hence, in (81), $V_{m+1}(n) = 0$ for $n < D^{10 K_{14}}$; $\delta n \geq n^{0{,}9}$ for $n \geq D^{10 K_{14}}$, $E(\delta n, m+1)$ and $E(\delta n, m+2)$ are $< \exp(-10^{-3} n^{0{,}9})$

$$\left|\sum_m\right| < 2\cdot 10^4 \cdot e^{-\frac{m}{2}} \cdot \sum_{n \geq D^{10K_{14}}} (\ln D)^{m+2} \exp(-10^{-3} n^{0{,}9}) <$$

$$< 2\cdot 10^4 \cdot e^{-\frac{m}{2}} \sum_{n \geq D^{10 K_{14}}} \exp(-10^{-3} n^{0{,}9} + 2 \ln D \ln\ln n) <$$

$$< 2\cdot 10^4 \cdot e^{-\frac{m}{2}} \sum_{n \geq D^{K_{14}\cdot 10}} \exp(-10^{-4} \cdot n^{0{,}9}) < \exp(-10^{-5} D).$$

Hence, the error in omitting $\sum_m$'s with $m \geq M_\Psi$, $m \leq \ln D$, in (80) cannot exceed $\frac{1}{(\ln D)^{M_\Psi/2}} \exp(-10^{-5} D)$ and we obtain:

$$|\Phi(x, \chi, Z, \tau_0, \tau_1)| \leq \sum_{m=0}^{M_\Psi} \frac{1}{(\ln D)^{m+1}} \left|\sum_m\right| + R_2 \tag{84}$$

with $|R_2| < 2\exp\left(-\frac{1}{2} \ln D \ln\ln D\right)$.

## § 26

We return now to (56), and wish to prove for $\Psi(D) > K_{11}$ the estimate analogous to (11):

Lemma XXIII. *For* $\Psi(D) > K_{11}$,

$$\int_{X_1}^{X_2} |\Phi(x, \chi, Z, \tau_0, \tau_1)|^2 \exp\left(2 \frac{\Psi(D)}{\ln D} - 2\right) x\, dx \geq \frac{\ln D}{4(\Psi(D))^2} \tag{85}$$

*with* $X_1$ *and* $X_2$ *those of Lemma XIX.*

Proof. We have by (55)

$$\Phi(x, \chi, Z, \tau_0, \tau_1) = S(x, \chi, Z, \tau_0, \tau_1) e^{\alpha_0 x} + R_{\mathfrak{S}},$$
$$|R_{\mathfrak{S}}| < \delta^{-0{,}66} = e^{-0{,}66x} \text{ for } x \in [X_1, X_2].$$

Hence

$$\Phi(x, \chi, Z, \tau_0, \tau_1) e^{-\alpha_0 x} = S(x, \chi, Z, \tau_0, \tau_1) + R_x, \quad |R_x| < e^{-0{,}2x}.$$

since $\alpha_0 \geqslant 1 - \frac{\Psi(D)}{\ln D} > 0{,}86$;

$$|S(x, \chi, Z, \tau_0, \tau_1)|^2 \leqslant 2|\Phi(x, \chi, Z, \tau_0, \tau_1)|^2 e^{-2\alpha_0 x} + 2R_x^2,$$

$$\int_{X_1}^{X_2} |S(x, \chi, Z, \tau_0, \tau_1)|^2 dx \leqslant 2\int_{X_1}^{X_2} |\Phi(x, \chi, Z, \tau_0, \tau_1)|^2 e^{-2\alpha_0 x} dx + 2\int_{X_1}^{X_2} e^{-0{,}4x} dx.$$

Hence

$$\int_{X_1}^{X_2} |\Phi(x, \chi, Z, \tau_0, \tau_1)|^2 e^{-2\alpha_0 x} dx \geqslant \frac{1}{2}\int_{X_1}^{X_2} |S(x, \chi, Z, \tau_0, \tau_1)|^2 dx - \frac{1}{D} > \frac{\ln D}{4(\Psi(D))^2}$$

by (56), since $\Psi(D) \leqslant (\ln D)^{0{,}001}$.

As $2\alpha_0 \geqslant 2\left(1 - \frac{\Psi(D)}{\ln D}\right)$ we have

$$e^{-2\alpha_0 x} \leqslant \exp\left(2\frac{\Psi(D)}{\ln D} - 2\right)x,$$

and hence we obtain (85).

## § 27

The estimates (84) and (85) hold for any $L(s, \chi)$ belonging to the case II of § 8, and having zeros in $\sigma \geqslant 1 - \frac{\Psi(D)}{\ln D}$; $|t| \leqslant (\Psi(D))^{100}$ with $\Psi(D) \in [K_{11}, (\ln D)^{0{,}001}]$. The right hand sides of the estimates (84) and (85) do not depend upon $Z$, but (84) depends upon $\tau_0$ and $\tau_1$ as they are involved in $E(\delta n, m+1)$ and $E(\delta n, m+2)$.

We must now recall the definition and formation of these numbers.

In § 10, Lemma VIII, we firstly constructed for any $L$-series of described type a real number $\eta_1$, with $|\eta_1| \leqslant (\Psi(D))^{75}$.

In § 12 for a fixed $\eta_1$ we chose for $\tau_0$ an arbitrary real number of the segment $[\eta_1 - 1, \eta_1 + 1]$

In § 16 was fixed a number $\nu$ with $|I\nu - \tau_0| \leqslant (\Psi(D))^{10}$.

In § 21 we chose for $\tau_0$ an arbitrary real number such that $|\tau_1 - I\nu| \leqslant 2$ and $|\tau_0 - \tau_1| \geqslant 0{,}1$.

Lemma XXIV. *If* $Q\left(\frac{\Psi(D)}{\ln D}\right) - Q_1\left(\frac{\Psi(D)}{\ln D}\right) = Q_2\left(\frac{\Psi(D)}{\ln D}\right) > \exp(10^6\Psi(D))$, *then there exist* $\geqslant \frac{1}{(\Psi(D))^{100}} Q_2\left(\frac{\Psi(D)}{\ln D}\right)$ *L-series for which the numbers* $\tau_0$ *and* $\tau_1$ *are the same for all series, so that* (84) *and* (85) *hold.*

Proof. We can obviously choose $\geqslant \frac{1}{(\Psi(D))^{80}} Q_2\left(\frac{\Psi(D)}{\ln D}\right)$ $L$-series for which the numbers $\eta_1 = \eta_1(\chi)$ differ no more than by 0,001. We consider now the segments $|t - \eta_1(\chi)| \leqslant 0{,}05$. Their intersection contains a segment of length $>0{,}8$ and we can fix a $\tau_0$ as any number of this intersection.

We now consider the corresponding number $\nu = \nu(\chi)$. As $|I\nu - \tau_0| \leqslant (\Psi(D))^{10}$, we can find $\geqslant \frac{1}{(\Psi(D))^{100}} Q_2\left(\frac{\Psi(D)}{\ln D}\right)$ series for which $I\nu(\chi)$

differ no more than by 0,001. The segments $|t-I\nu(\chi)|\leqslant 1$ have the intersection, containing a segment of length $>1$; $\tau_1$ may be an arbitrary number of this segment, and we fix it such that $|\tau_0-\tau_1|\geqslant 0,1$. Thus, the lemma is proved.

## § 28

We denote now by $Q_3\left(\frac{\Psi(D)}{\ln D}\right)$ the quantity of our $L$-series of Lemma XXIV, for which $\tau_0$ and $\tau_1$ are the same. We have then:

$$\left.\begin{aligned} &\text{either } Q_2\left(\frac{\Psi(D)}{\ln D}\right)<\exp\left(10^6\,\Psi(D)\right)\\ &\text{or } Q_3\left(\frac{\Psi(D)}{\ln D}\right)>\frac{1}{(\Psi(D))^{100}}\,Q_2\left(\frac{\Psi(D)}{\ln D}\right).\end{aligned}\right\} \tag{86}$$

Let $\chi_{\alpha_1},\chi_{\alpha_2},\ldots,\chi_{\alpha_{Q_3}}$ be the characters of our $L$-series. Then we have obviously by (85)

$$\int\limits_{X_1}^{X_2}\sum_{j=1}^{Q_3}|\Phi(x,\chi_{\alpha_j},Z_j,\tau_0,\tau_1)|^2\exp\left(2\frac{\Psi(D)}{\ln D}-2\right)x\,dx\geqslant Q_3\left(\frac{\Psi(D)}{\ln D}\right)\cdot\frac{\ln D}{4(\Psi(D))^2} \tag{87}$$

($Z_j$ is a complex number depending on $j$).

We return now to (84) and prove the following

Lemma XXV:

$$\sum_{\chi}\sum_{m=0}^{M_\Psi}\left\{\frac{1}{(\ln D)^{m+1}}\left|\sum\nolimits_m\right|^2\right\}\leqslant\delta^{-2}\exp\left(K_5\,\Psi(D)\right), \tag{88}$$

*where the summation is extended over all possible $\chi$'s including the principal one, and*

$$-\ln\delta=x\in[X_1,X_2].$$

Proof. For a given $m\leqslant M_\Psi$ we have, by orthogonality of the $\chi$'s

$$\sum_{\chi}\left|\sum_{n=2}^{\infty}\chi(n)\frac{V_{m+1}(n)}{(\ln D)^{m+1}}\left[\frac{\ln n}{m+1}E(\delta n,m+1)+E(\delta n,m+2)\right]\right|^2=$$

$$=\varphi(D)\sum_{n=2}^{\infty}\left\{\frac{V_{m+1}(n)}{(\ln D)^{m+1}}\left[\frac{\ln n}{m+1}E(\delta n,m+1)+E(\delta n,m+2)\right]\cdot\right.$$

$$\left.\cdot\sum_{n_1\equiv n\ (\mathrm{mod}\ D)}\frac{V_{m+1}(n_1)}{(\ln D)^{m+1}}\left[\frac{\ln n_1}{m+1}\overline{E(\delta n_1,m+1)}+\overline{E(\delta n_1,m+2)}\right]\right\}.$$

Here $\varphi(D)$ denotes Euler's function, the summation over $n_1$ is extended over all natural numbers $n_1$ congruous to $n\ (\mathrm{mod}\ D)$ and the horizontal bars denote conjugate values.

We consider firstly for a fixed $n$ the sum

$$\sum_{n_1\equiv n\ (\mathrm{mod}\ D)}\frac{V_{m+1}(n_1)}{(\ln D)^{m+1}}\left[\frac{\ln n_1}{m+1}\cdot\overline{E(\delta n_1,m+1)}+\overline{E(\delta n_1,m+2)}\right].$$

For $\delta n_1 \geqslant D$, $n_1 \geqslant D\delta^{-1}$, we have by (78) and (74), that the corresponding terms may be omitted with an error $< \frac{1}{D}$. Hence the modulus of our sum cannot exceed

$$\sum_{\substack{n_1 \equiv n (\mathrm{mod} D) \\ n_1 \leqslant D\delta^{-1}}} \frac{V_{m+1}(n_1)}{(\ln D)^{m+1}} \left[\frac{\ln n_1}{m+1} |E(\delta n_1, m+1)| + |E(\delta n_1, m+2)|\right] + 1.$$

Now, taking into account (74) and (76) we see that the sum of terms with $n_1 \leqslant \delta^{-1/2}$ does not exceed

$$\delta^{-1/2} \cdot K_4^{M_\Psi} \cdot |\ln \delta|^{M_\Psi} < \delta^{-0,6}. \tag{89}$$

We consider now the interval $\delta^{-1/2} \leqslant n_1 \leqslant \delta^{-1}$. We divide it into partial intervals of the type $\left[\frac{\delta^{-1}}{2^{r+1}}, \frac{\delta^{-1}}{2^r}\right]$; the extreme left interval may project out of our segment, but it can be neglected with an error $< \delta^{-0,6}$. In the segment $\left[\frac{\delta^{-1}}{2^{r+1}}, \frac{\delta^{-1}}{2^r}\right]$ we shall have: $|\ln \delta n_1| < (r+1) \ln 2$, and by (76)

$$|E(\delta n_1, m+1)| + |E(\delta n_1, m+2)| < \exp(2m_1 \ln(r+1)).$$

Here $m_1 = m+2 \leqslant 2M_\Psi = 2 \cdot 10^4 \cdot K_{14} \cdot \frac{\Psi(D)}{\ln \Psi(D)}$.

Hence it is easy to see that

$$|E(\delta n_1, m+1)| + |E(\delta n_1, m+2)| \leqslant \exp K_{16} \cdot \Psi(D) \cdot 2^{\frac{r+1}{2}}. \tag{90}$$

In fact, if $2^{\frac{r+1}{2}} < \exp(2m_1 \ln(r+1))$, then $\frac{r+1}{\ln(r+1)} < 100\, m_1$ and hence

$$\ln(r+1) < \ln m_1 + C_{28},$$

$$\exp(2m_1 \ln(r+1)) < \exp(2m_1 \ln m_1 + C_{28} m_1) <$$
$$< \exp(C_{29} M_\Psi \ln M_\Psi) < \exp(K_{16} \Psi(D)).$$

We now obtain by (74):

$$\frac{V_{m+1}(n_1)}{(\ln D)^{m+1}} \ln n < C_{30} K_{14}^{m+2} \ln D < \ln D \cdot \exp(K_{17} \Psi(D)). \tag{91}$$

Hence, the partial sum for $n_1 \in \left[\frac{\delta^{-1}}{2^{r+1}}, \frac{\delta^{-1}}{2^r}\right]$ cannot exceed $2^{\frac{r+1}{2}} \cdot$ $\cdot \ln D \cdot \exp(K_{18} \Psi(D))$ multiplied into the quantity of the numbers $n_1$, belonging to that segment, congruous to a fixed $n$ and having only the prime factors $\geqslant N_\Psi$. We thus arrive at the necessity of using the generalised Brun—Titchmarsh theorem (cf. § 9, Lemma VII). The trivial estimate is in this case $\frac{1}{2^r} \cdot \frac{\delta^{-1}}{D}$. This trivial estimate is sufficient for $\Psi(D) > \ln \ln D$ (the case of the «middle stripe» in § 3), since then $\exp(A\Psi(D)) \geqslant (\ln D)^A$ and the adding or deleting such factors as $\ln D$ cannot change the situation.

But it is not so for $\Psi(D) \leqslant \ln \ln D$, $\Psi(D) > K_{11}$ («Viggo—Brun's stripe»). Here we need the following fundamental

L e m m a XXVI. *Suppose that we are given a segment of an arithmetic progression*

$$1 \leqslant Dx + n \leqslant N \quad \textit{with} \quad N \geqslant D^2.$$

*Let $N_\Psi$ be a number under conditions*

$$2 \leqslant N_\Psi \leqslant N^{0,001}.$$

*Then the quantity of numbers $n_1$ belonging to our segment of progression, and having only prime factors $\geqslant N_\Psi$ does not exceed*

$$C_{31} \frac{N}{\varphi(D) \ln N_\Psi}. \tag{92}$$

Proof. This is A. A. Buchstab's generalization of Brun—Titchmarsh's theorem (Lemma VII). It is proved by the Viggo Brun's sieve method without any essential additions.

Returning to the segment $\left[\frac{\delta^{-1}}{2^{r+1}}, \frac{\delta^{-1}}{2^r}\right]$, we note that $\delta^{-1}/2^{r+1} > \frac{1}{2}\delta^{-0,5} > \frac{e^{0,5X_1}}{2} > \frac{D^4}{2}$, and $N_\Psi = \exp\left(0,001 \frac{\ln D}{\Psi(D)} \cdot \ln \Psi(D)\right) < N^{0,001}$, so that we may apply the lemma, and, combining it with (90) and (91), we obtain the estimate:

$$2^{\frac{r+1}{2}} \ln D \cdot \exp(K_{18}\Psi(D)) \cdot \frac{\delta^{-1}}{2^r} \cdot C_{31} \cdot \frac{1}{\varphi(D) \ln N_\Psi} <$$
$$< 2C_{31} \cdot \frac{\delta^{-1}}{2^{r/2}} \cdot \frac{1}{\varphi(D)} \cdot \exp(K_{18}\Psi(D)) \cdot \frac{\ln D \cdot \Psi(D)}{0,001 \ln D \cdot \ln \Psi(D)} <$$
$$< C_{32} \frac{\delta^{-1}}{\varphi(D)} \cdot \frac{1}{2^{r/2}} \cdot \exp(2K_{18}\Psi(D)). \tag{93}$$

Summing up these estimates for $r = 1, 2, \ldots$ and taking into account that $\delta^{-1}/\varphi(D) > \delta^{-0,6}$, we obtain

$$\sum_{\substack{n_1 \equiv n \pmod D \\ n_1 \leqslant \delta^{-1}}} \frac{V_{m+1}(n_1)}{(\ln D)^{m+1}} \cdot \left[\frac{\ln n}{m+1} |E(\delta n_1, m+1)| + |E(\delta n_1, m+2)|\right] \leqslant$$
$$\leqslant C_{33} \frac{\delta^{-1}}{\varphi(D)} \exp(2K_{18}\Psi(D)).$$

It remains now the segment $n_1 \in [\delta^{-1}, D\delta^{-1}]$. We divide it into partial segments of the type $[\delta^{-1} \cdot 2^r, \delta^{-1} \cdot 2^{r+1}]$. Here the last segment may be dropped because of the rapid decrease of $E(\delta n, m+1)$ and $E(\delta n, m+2)$ by (78). Now, for $n_1$ in this segment we have, by (77),

$$\delta n_1 > 2^r, \qquad |E(\delta n_1, m+1)| + |E(\delta n_1, m+2)| <$$
$$< \exp\left(-\frac{2^r}{4}\right) \exp(C_{26}(m+2)\ln(m+2)) < \exp\left(-\frac{2^r}{4}\right) \exp(K_{19}\Psi(D)).$$

This is the only difference with the preceding estimate, and so we obtain:

$$\sum_{\substack{n_1 \equiv n \pmod D \\ \delta^{-1} \leqslant n_1 \leqslant D\delta^{-1}}} \frac{V_{m+1}(n)}{(\ln D)^{m+1}} \left[\frac{\ln n}{m+1} \cdot |E(\delta n, m+1)| + |E(\delta n, m+2)|\right] \leqslant$$
$$\leqslant \frac{\delta^{-1}}{\varphi(D)} \cdot \exp[K_{21}\Psi(D)]. \tag{94}$$

Hence the whole sum

$$\left|\sum_{n_1 \equiv n \pmod D} \frac{V_{m+1}(n_1)}{(\ln D)^{m+1}} \left[\frac{\ln n_1}{m+1} \overline{E(\delta n_1, m+1)} + \overline{E(\delta n_1, m+2)}\right]\right| \leqslant$$
$$\leqslant \frac{\delta^{-1}}{\varphi(D)} \cdot \exp(K_{21}\Psi(D)) \tag{95}$$

and hence

$$\sum_{\chi}\left|\frac{\sum_m}{(\ln D)^{m+1}}\right|^2 \leqslant \varphi(D)\cdot\frac{\delta^{-1}}{\varphi(D)}\cdot\exp(K_{21}\Psi(D))\cdot$$

$$\cdot\sum_{n=2}^{\infty}\frac{V_{m+1}(n)}{(\ln D)^{m+1}}\left[\frac{\ln n}{m+1}\cdot|E(\delta n, m+1)|+|E(\delta n, m+2)|\right].$$

The last sum is of the type considered above, but with $D=1$, so that it cannot exceed $\delta^{-1}\cdot\exp(K_{21}\Psi(D))$ and thus

$$\sum_{\chi}\left|\frac{\sum_m}{(\ln D)^{m+1}}\right|^2 \leqslant \delta^{-2}\cdot\exp[2K_{21}\Psi(D)].$$

As the quantity of $\sum_m$'s is $M_\Psi < \exp(K_{14}\Psi(D))$, we obtain (88), q. e. d.

### § 29. Proof of the basic theorem

We are now able to prove the basic theorem (3):

$$Q\left(\frac{\Psi(D)}{\ln D}\right) \leqslant \exp(A\Psi(D))$$

for $\Psi(D)\in[2, (\ln D)^{0,001}]$, and, consequently, for $\Psi(D)\in\left[2, \frac{1}{3}\ln D\right]$, i. e. in its full extent (cf. § 4).

As $Q\left(\frac{\Psi(D)}{\ln D}\right)$ is a non-decreasing function, by its definition, we can consider $\Psi(D)\geqslant K_{11}$. Such a value of $\Psi(D)$ being fixed, we have by (86): either $Q_2\left(\frac{\Psi(D)}{\ln D}\right) < \exp(10^6\,\Psi(D))$, or (86) holds. In the first case, (3) is proved, by lemma VI.

In the second case we consider the corresponding characters $\chi_{\alpha_1}, \chi_{\alpha_2}, \ldots$ $\ldots, \chi_{\alpha_{Q_2}}$. For each one of them we have, by (64),

$$|\Phi(x, \chi_{\alpha_j}, Z_j, \tau_0, \tau_1)| \leqslant \sum_{m=0}^{M_\Psi}\frac{1}{(\ln D)^{m+1}}\left|\sum\nolimits_m\right| + 1.$$

Hence, by Schwarz's inequality:

$$|\Phi(x, \chi_{\alpha_j}, Z_j, \tau_0, \tau_1)| \leqslant 2M_\Psi\sum_{m=0}^{M_\Psi}\left|\frac{\sum_m}{(\ln D)^{m+1}}\right|^2 + 2M_\Psi.$$

Hence

$$\sum_{j=1}^{Q}|\Phi(x, \chi_{\alpha_j}, Z_j, \tau_0, \tau_1)|^2 \leqslant 2M_\Psi\sum_{\chi_j}\sum_{m=0}^{M_\Psi}\left\{\frac{1}{(\ln D)^{m+1}}\left|\sum\nolimits_m\right|\right\}^2 + 2M_\Psi D.$$

The last sum cannot exceed the sum extended over all possible $\chi$'s (mod $D$), and so, by Lemma XXV (88)

$$\sum_{j=1}^{Q_2}|\Phi(x, \chi_{\alpha_j}, Z_j, \tau_0, \tau_1)|^2 \leqslant 2M_\Psi\cdot\delta^{-2}\cdot\exp(K_{15}\Psi(D)) + 2M_\Psi D \leqslant$$

$$\leqslant 2e^{2x}\cdot\exp(2K_{15}\Psi(D)) \tag{96}$$

as $x=-\ln\delta$, $\delta^{-2}\geqslant D^8$, $M_\Psi=10^4K_{14}\cdot\frac{\Psi(D)}{\ln\Psi(D)}$. Substituting the right hand side of (95) into the left hand side of (87), we replace the integrand by the larger quantity (95), and so we shall have *a fortiori*

$$\int_{X_1}^{X_2} 2e^{2x}\cdot\exp(2K_{15}\Psi(D))\cdot\exp\left(2\frac{\Psi(D)}{\ln D}-2\right)x\,dx\geqslant Q_3\left(\frac{\Psi(D)}{\ln D}\right)\cdot\frac{\ln D}{4(\Psi(D))^2}.$$

The left hand side cannot exceed:

$$2\exp(2K_{15}\Psi(D))\cdot\frac{\ln D}{2\Psi(D)}\cdot\exp\left(2\frac{\Psi(D)}{\ln D}\cdot X_2\right)\leqslant\frac{\ln D}{\Psi(D)}\cdot\exp(K_{22}\Psi(D)),$$

since $X_2=K_{13}\ln D$. Hence

$$Q_3\left(\frac{\Psi(D)}{\ln D}\right)\cdot\frac{\ln D}{4(\Psi(D))^2}\leqslant\frac{\ln D}{\Psi(D)}\cdot\exp(K_{22}\Psi(D)),$$

$$Q_3\left(\frac{\Psi(D)}{\ln D}\right)\leqslant 4\Psi(D)\cdot\exp(K_{22}\Psi(D)).$$

Hence, by (86),

$$Q_2\left(\frac{\ln D}{\Psi(D)}\right)\leqslant 4(\Psi(D))^{101}\cdot\exp(K_{22}\Psi(D))\leqslant\exp(K_{22}\Psi(D)).$$

Taking into account (6), Lemma VI, we have proved the basic theorem (3) with

### § 30. Proof of the relation $p_{\min}(l, D)<D^{C_0}$ for one half of progressions (mod $D$)

The basic theorem, without additions about «Deuring-Heilbronn effect» enables us to prove the relation (2) for one half of $l$'s (mod $D$).

We consider now the rectangle $|t|\leqslant D$ of «Siegel's stripe» $1-\frac{c_0}{\ln D}\leqslant\sigma\leqslant 1$. By a remarkable theorem by Landau and Page (cf. Page, [5], Lemma 9) among our $L(s,\chi)$'s there can be only one series $L(s, \mathrm{X})$ corresponding to a non-principal real character $\mathrm{X}$ primitive or non primitive, such that $L(s,\mathrm{X})$ may have zeros in that rectangle; in this case, it has only one real zero $\beta_0$. We denote now by $\mathfrak{G}_D$ the system of all integer numbers $l$ of the segment $[1, D-1]$ such that $\mathrm{X}(l)=-1$. We shall prove the

Theorem. *There exists an absolute constant $C_0$ such that if $l\in\mathfrak{G}_D$, then the least prime in the arithmetic progression $Dx+l$ satisfies the inequality*

$$p_{\min}(l, D)\leqslant D^{C_0}. \tag{97}$$

*Moreover, if*

$$S(N, l)=\sum_{n\equiv l\,(\mathrm{mod}\,D)}\Lambda(n)e^{-n/N},\quad x=\ln N,\quad (l, D)=1,$$

*then*

$$\int_M^{2M}dx\int_x^{2x}\frac{S(N,l)}{N}dx=\frac{1}{\varphi(D)}\cdot\frac{3M^2}{2}\left(1+2\theta\cdot\exp\left(-c_1\frac{M}{\ln D}\right)+\frac{\theta}{\ln D}\right)-$$

$$-\frac{\mathrm{X}(l)}{\varphi(D)}\Gamma(\beta_0)\frac{P(e^{-\sigma_0 M})}{\sigma_0^2}, \tag{98}$$

*where* $\sigma_0 = 1 - \beta_0$, $P(y) = \frac{y(y-1)^2(y+2)}{2}$, *and the last term should be dropped if* X *does not exist. M is assumed* $> B_2 \ln D$.

Corollary. *If* $D \equiv 1 \pmod 4$ *is a prime, and* $\left(\frac{l}{D}\right) = -1$, *then* (97) *holds.*

Proof of (98). We divide the stripe $\frac{3}{4} \leqslant \sigma \leqslant 1$ into the stripes $\mathfrak{B}_k$:

$$1 - \frac{2^{k+1}c_0}{\ln D} \leqslant \sigma \leqslant 1 - \frac{2^k c_0}{\ln D} \quad (k = 0, 1, 2, \ldots, b)$$

$b$ being the least integer for which $2^b c_0 \geqslant \frac{1}{3} \ln D$. We cut out of each stripe belonging to the number $k$ the rectangle $|t| \leqslant \min(2^{100(k+2)}, \ln^3 D)$ and denote it by $R_{k_0}$. Let $Q_k$ be the quantity of $L$-series $L(s, \chi)$ with zeros in $R_{k_0}$. We derive easily from the basic theorem (3):

$$Q_k \leqslant \exp(B \cdot 2^k)$$

$B$ being an absolute constant; $Q_0 \leqslant 1$. We have now [1]

$$\varphi(D) \cdot S(N, l) = \sum_{n \equiv l \pmod D}^{1, \ldots, \infty} \Lambda(n) e^{-n/N} =$$

$$= N - \mathrm{X}(l)\Gamma(\beta_0) \cdot N^{\beta_0} - \sum_{k=1}^{l} Z_k + \theta \cdot \frac{N}{\ln D} \quad (N > D^{10}), \tag{99}$$

where

$$Z_k = \sum_{\chi} \frac{1}{\chi(l)} \sum_{\rho_j} \Gamma(\rho_j) N^{\rho_j}$$

the inner sum being extended over the zeros of $L(s, \chi)$ in the stripe $\mathfrak{B}_k$ and the outer one over all $\chi$'s such that $L(s, \chi)$ has zeros in $\mathfrak{B}_k$.

We cut out of $\mathfrak{B}_k$ $(k \geqslant 1)$ a sequence of pairs of symmetrical rectangles $R_{k_0}$;

$$2^{k \cdot 100} \leqslant |t| \leqslant 2^{k \cdot 200},\ 2^{k \cdot 200} \leqslant |t| \leqslant 2^{k \cdot 300}, \ldots,\ 2^{k \cdot g_k \cdot 100} \leqslant |t| \leqslant \ln^2 D,$$

and denote then by $R_{k_0}, R_{k_1}, \ldots, R_{k g_k}$. Let $Q_{kn}$ be the number of $L$-series with zeros in $R_{kn}$. Then, by definition of $Q\left(\frac{\Psi(D)}{\ln D}\right)$, we shall have:

$$Q_{kn} \leqslant Q\left(\frac{2^{kn}}{\ln D}\right) \leqslant \exp(B \cdot 2^{k(n+1)}). \tag{100}$$

Dividing the equation for $\varphi(D)$ by $N$ and denoting $\rho_j = \beta_j + it_j$, $\rho_j - 1 = -\sigma_j + it_j$, we obtain:

$$\frac{\varphi(D)}{N} S(N, l) = 1 + \mathrm{X}(l)\Gamma(\beta_0)\exp(-\sigma_0 x) + \sum_{k=1}^{b} \frac{Z_k}{N} + \frac{\theta}{\ln D} \tag{101}$$

(since $x = \ln N$).

We consider now $Z_k$'s with $2^{k-1}c_0 > \ln\ln D$. For $N > D^{10}$ we have by, virtue of Lemma IV,

$$\left|\frac{Z_k}{N}\right| \leqslant C_{34} \ln D \cdot \exp\left(-x \cdot c_0 \cdot \frac{2^{k-1}}{\ln D}\right) \cdot \sum_{n=0}^{g_k} \exp(B \cdot 2^{k(n+1)}) \cdot$$

$$\cdot \exp(-2^{100kn}) \leqslant C_{35} \ln D \cdot \exp(B \cdot 2^k) \cdot \exp\left(-x \frac{2^{k-1}c_0}{\ln D}\right) < \frac{1}{\ln^2 D} \cdot \frac{1}{2^{k+1}},$$

for $x > 10^3 B \ln D = B_1 \ln D$; hence for such $x$'s and $2^{k-1} \geqslant \ln\ln D$ we have

$$\sum_{2^{k-1}c_0 \geqslant \ln\ln D} \left|\frac{Z_k}{N}\right| < \frac{1}{\ln^2 D}$$

and therefore

$$\frac{\varphi(D)}{N}\cdot S(N, l) = 1 + X(l)\Gamma(\beta_0)\exp(-\sigma_0 x) + \sum_{k=1}^{4c_0 \ln\ln\ln D} \frac{Z_k}{N} + \frac{2\theta}{\ln D}\cdot \qquad (102)$$

For a fixed $k$ each particular term of $\frac{Z_k}{N}$ is of the form

$$\frac{1}{\chi(l)}\Gamma(\rho_j)\exp(-\sigma_j + it_j)x.$$

We form now

$$\int_x^{2x} \frac{Z_k}{N}\,dx = \sum_\chi \frac{1}{\chi(l)} \sum_{\rho_j} \Gamma(\rho_j) \left\{ \frac{\exp(-\sigma_j + it_j)2x}{-\sigma_j + it_j} - \frac{\exp(-\sigma_j + it_j)x}{-\sigma_j + it_j} \right\}$$

and then

$$\int_M^{2M} dx \int_x^{2x} \frac{Z_k}{N}\,dx = \sum_\chi \frac{1}{\chi(l)} \sum_{\rho_j} \Gamma(\rho_j)\cdot$$

$$\cdot\left\{ \frac{\exp(-\sigma_j + it_j)\cdot 4M}{2(-\sigma_j + it_j)^2} - \frac{\exp(-\sigma_j + it_j)\cdot 2M}{2(-\sigma_j + it_j)^2} - \right.$$
$$\left. - \frac{\exp(-\sigma_j + it_j)\cdot 2M}{(-\sigma_j + it_j)^2} + \frac{\exp(-\sigma_j + it_j)\cdot M}{(-\sigma_j + it_j)^2} \right\}, \quad M > B_1 \ln D.$$

Now, taking into account (100) and the Lemma IV, we obtain

$$\left|\int_M^{2M} dx \int_x^{2x} \frac{Z_k}{N}\,dx\right| < \exp\left(-2^{k-1}c_0 \frac{M}{\ln D}\right)\cdot$$

$$\cdot \sum_{n=0}^{\rho_k} \left\{ \exp(B\cdot 2^{k(n+1)})\cdot \exp(-2^{100kn})\cdot \sum \frac{C_{35}\cdot 2^m \cdot \ln^2 D}{c_0^2\cdot 2^{2m}} \right\} <$$

$$< \frac{\ln^2 D}{2^{k+1}} \exp\left(-\frac{c_0}{2}\frac{M}{\ln D}\right).$$

Hence, substituting into (102) and summing up, we get

$$\int_M^{2M} dx \int_x^{2x} \varphi(D) e^{-x} S(e^x, l)\,dx = \int_M^{2M} dx \int_x^{2x} dx -$$
$$- X(l)\Gamma(\beta_0) \int_M^{2M} dx \int_x^{2x} e^{-\sigma_0 x}\,dx + \frac{\theta}{\ln D}\int_M^{2M} dx \int_x^{2x} dx +$$
$$+ \theta\cdot \ln^2 D\cdot \exp\left(-\frac{c_0}{2}\frac{M}{\ln D}\right) = \frac{3M^2}{2}\left(1 + 2\theta\cdot\exp\left(-\frac{c_0}{2}\frac{M}{\ln D}\right) + \frac{\theta}{\ln D}\right) -$$
$$- X(l)\Gamma(\beta_0)\frac{P(e^{-\sigma_0}M)}{\sigma_0^2},$$

q. e. d.

Proof of the corollary. If $D$ is a prime, $X$ is primitive. By a lemma by A. Walfisz [8], in this case $X(n) = \left(\frac{\pm D}{n}\right)$, where $\pm D$ is a fundamental discriminant. If $D \equiv 1 \pmod 4$, the sign $(+)$ is to be taken, and, by the law of reciprocity,

$$X(n) = \left(\frac{D}{n}\right) = \left(\frac{n}{D}\right).$$

Hence, $X(l) = -1$ for $\left(\frac{l}{D}\right) = -1$, and (97) follows from (98).

The sections II and III are to be published in this journal.

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# О наименьшем простом числе в арифметической прогрессии
## I. Основная теорема

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*(Резюме)*

В настоящей статье дается способ устранить недоказанное предположение, применявшееся автором в заметке «О возможности обойти расширенную гипотезу Riemann'а». В остальном же схема рассуждений такая же, как и там. Доказывается

Теорема. *Если* $L(s, \chi_1)$, $L(s, \chi_2)$, $\ldots$, $L(s, \chi_{\varphi(D)})$ *суть все* $L$-*ряды* $(\mathrm{mod}\, D)$ *и* $\Psi(D) \in \left[2, \frac{1}{3}\ln D\right]$, *то*

$$Q\left(\frac{\Psi(D)}{\ln D}\right) \leqslant \exp A\,\Psi(D), \tag{1}$$

*где слева количество* $L$-*рядов, имеющих хоть один нуль в прямоугольнике* $1 \geqslant \sigma \geqslant 1 - \frac{\Psi(D)}{\ln D}$, $|t| \leqslant \{(\Psi(D))^{100}, \ln^3 D\}$, $A > 0$ — *абсолютная константа.*

Из дополненной «эффектом Deuring — Heilbronn'а», изложенным во второй части статьи, теоремы вытекает, что если $p_{\min}(l, D)$ — минимальное простое число в прогрессии $Dx + l$; $l \in [1, D-1]$; $(l, D) = 1$, то

$$p_{\min}(l, D) < D^{C_0} \quad (C_0 \text{ абс. константа}). \tag{2}$$

Сама по себе теорема (без дополнения во второй части) достаточна лишь для доказательства (2) при $D \equiv 1 \pmod 4$ простом и $\left(\frac{l}{D}\right) = -1$.

Из нее следует также некоторая асимптотическая закономерность в распределении малых простых чисел в прогрессии.

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